Online Appendices for:

"Estimating Intergenerational Returns to Medical Care: New Evidence from At-Risk Newborns"

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Table of Contents

A Data Appendix **A1**

A.1 Matching Microdata A1

A.2 Descriptive Figures and Tables A2

B Identification Checks **B1**

B.1 Proof of Treatment B1

B.2 Identification Checks B2

C Additional Model Details **C1**

C.1 A Single Generation Model C1

C.2 Greater Granularity in Fertility Choices C1

C.3 Dynamic Health Stocks C2

C.4 Proceeding without a Normality Assumption C3

D Robustness Checks **D1**

E Additional Details: Selection into Treatment **E1**

E.1 Selection into Treatment in the First Generation E1

E.2 Compositional Effects and Fertility Selection E2

E.3 Additional Details: Marginal versus Average Effects in an RDD with Selection E2

F A Back-of-the-Envelope Estimation of Policy Costs and Benefits **F1**

A Data Appendix

In this appendix we collect additional information related to data and descriptive statistics referred to in the paper. Sub-section A.1 notes further information related to data matching in administrative datasets, and A.2 collects all descriptive figures in the order in which they are referred to in the text.

A.1 Matching Microdata

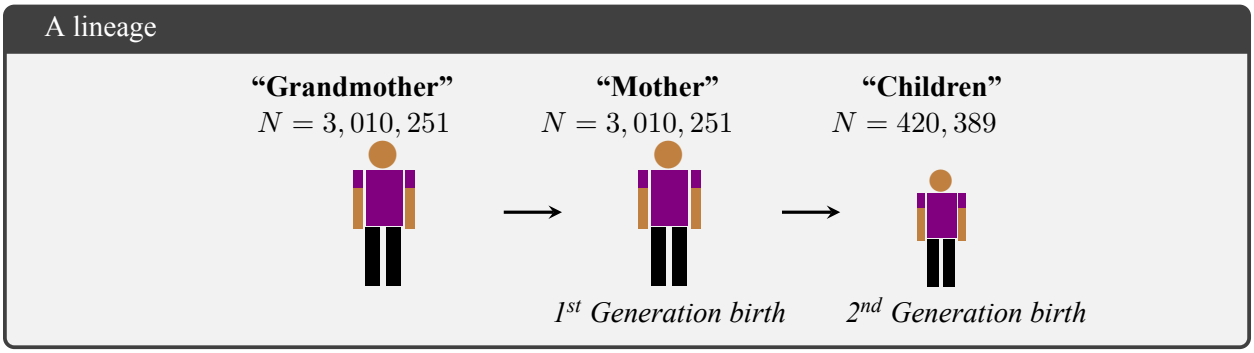
While individual microdata files are high quality and comprehensive, matching between microdata files requires access to restricted information—namely each individual's RUT, or national identity number. Fortunately, for a number of years Chile's Ministry of Health provided access to a unique anonymised version of this RUT which were harmonised across administrative records held by the Ministry of Health. This allows for us to match all administrative records in this paper (namely births, deaths, and inpatient hospitalisations) across files such that we generate a complete history for each individual. We can do this for the period of 1992-2018. This anonymised version of the RUT is generated by the Ministry of Health to match across *their own* administrative records, and we use this to generate the full data used in this paper. Nevertheless, the use of these matched microdata files implies two specific limitations which we lay out below.

Limitation 1: Matching Data with Administrative Data Held by Other Ministries The anonymised version of the RUT does not allow us to match health records with data held by other government authorities. Thus, we cannot observe information on later educational attainment for all births (which requires links to administrative records held by the Ministry of Education), nor their eventual partnership choices (which requires links to administrative records held by the Civil Registry). Such intra-agency data collaborations require formal collaboration agreements, and resultingly matched microdata cannot be shared. One benefit of using the anonymised version of the RUT generated by the Ministry of Health is that all matched data can be publicly available given that (i) no sensitive information is revealed in this process and (ii) all data and variables were originally shared publicly on their open data page (https://deis.minsal.cl/#datosabiertos) and are stored on internet archives such as the Wayback Machine (see, e.g. https://web.archive.org/web/20190326212649/https://deis.minsal.cl/#datosabiertos).

Limitation 2: Temporal Access 1992–2018 A second limitation relates to the period which we can study. In approximately 2020, the Ministry of Health changed their data sharing protocols and no longer publish these administrative databases with anonymised versions of the RUT. Because these microdata files are shared with around 1 year's delay, publicly shared data from 2019 onwards does not include an anonymised version of the RUT, meaning that we cannot match mothers from 2019 onwards with their own birth weights when they were born. Indeed, more generally the data sharing standards of the Ministry became more restrictive, and birth weight itself is no longer reported exactly, but rather reported at the individual level in categorical groups. Thus, we are limited with our analysis to consider periods from 1992–2018, implying that we can follow women who have turned a (maximum) of 26 years. Given that we observe no individuals who have completed their 26th of life, we limit our sample to mothers aged up to 25. This results in the removal of a small number of individuals who have turned 26 though not yet reached the end of their 26th year, though our results are not sensitive to this choice.

A.2 Descriptive Figures and Tables

Figure A1: Schematic Representation of a Lineage



Notes: Birth certificates capture characteristics of the individual born, as well as their parents, and we observe the unique identity number of the parent as well as the individual born. Thus, for "Mothers", at their time of birth we observe their birth outcomes, as well as their mothers' characteristics ("Grandmothers"). While for "Children", we observe their birth outcomes, their mother's birth outcomes, and their mother's outcomes at the time the child was born. The nomenclature "mother" refers to all girls who could potentially go on to being mothers, many of which have not yet given birth. Observations refer to quantities used in final estimation sample of mother's aged 15-45 at birth and without missing observations in covariates

Table A1: Matched Observations between Microdata Registers

Register	Observations with Valid Unique IDs	Matched to Births	Matched to Hospitalisation	Matched to Deaths
1992–2018 Births	6,617,637	435,013	5,656,409	83,841
2001–2019 hospitalisation	27,995,452	2,924,795	—	1,272,340
1992–2018 Deaths	2,393,583	83,841	4,627,999	—

Notes: Column 1 presents the total number of valid observations in each dataset. Column 2 notes the number of births which match to each dataset. In the case of the birth register, it refers to the number of births which match to other births in the data (*i.e.* mother–child links). Column 3 notes the number of hospitalisations which match to each other database. Note that in the case of births, the number of hospitalisations linked to births is not the same as the number of births linked to hospitalisations in the preceding column given that a single birth can be hospitalised multiple times. Finally, column 4 notes the total number of deaths which are matched with births occurring in the sample.

Table A2: Temporal Links between Mother–Child Matched Birth Years

Mother	Child														
Year	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
1992	10	57	346	1,475	3,789	6,205	7,965	9,468	9,964	9,898	9,831	9,438	8,954	8,663	8,912
1993	0	6	60	399	1,622	3,806	5,869	7,632	8,962	9,344	9,826	9,413	8,578	8,162	8,260
1994	0	0	7	60	373	1,614	3,608	5,982	7,429	8,457	9,273	9,023	8,519	7,974	7,865
1995	0	0	1	11	76	418	1,437	3,498	5,560	6,654	8,000	8,298	7,770	7,480	7,235
1996	0	0	0	3	9	83	356	1,484	3,493	4,967	6,205	7,046	7,305	7,068	6,774
1997	0	0	0	0	0	10	61	400	1,442	3,183	4,684	5,433	5,687	6,123	6,478
1998	0	0	0	0	0	0	5	56	367	1,431	2,911	3,882	4,366	4,899	5,566
1999	0	0	0	0	0	0	2	7	52	369	1,331	2,349	2,914	3,415	4,081
2000	0	0	0	0	0	0	0	2	6	56	326	1,057	1,879	2,270	2,736
2001	0	0	0	0	0	0	0	0	0	3	49	261	754	1,297	1,638
2002	0	0	0	0	0	0	0	0	0	0	4	43	248	582	1,022
2003	0	0	0	0	0	0	0	0	0	0	1	7	23	166	471
2004	0	0	0	0	0	0	0	0	0	0	0	1	6	48	177
2005	0	0	0	0	0	0	0	0	0	0	0	0	1	4	28
2006	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3

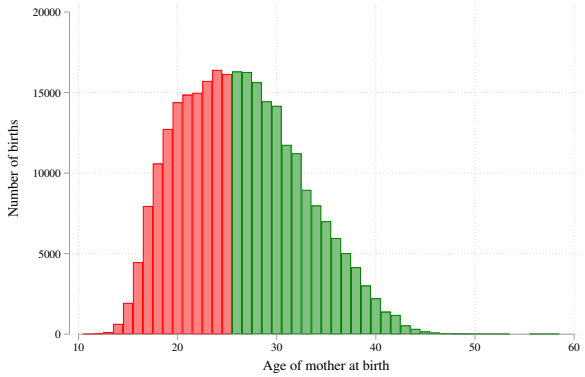
Notes: Values note the total number of mothers who are born in each of the years indicated in the left-hand column giving birth in each year indicated by row headers. All values are based on matched administrative records, and are provided from 1992 (the first birth year observed) up to 2006 (the last birth cohort in which a later-born child was linked to a mother born in this year.) A small number of observations (11) with inconsistencies in mother's age are removed prior to estimation.

Table A3: Summary Statistics – Births local to the 1,500 gram threshold

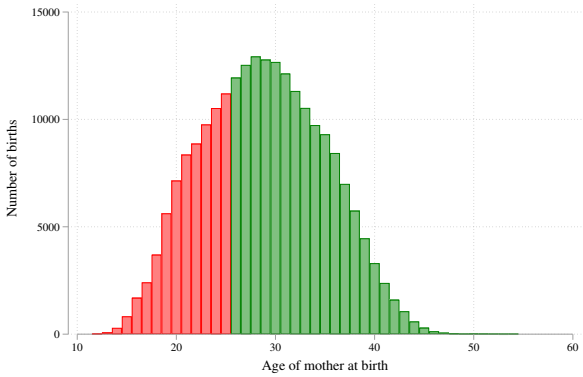
	Obs.	Mean	Std. Dev.	Min.	Max.
Panel A: First Generation Births					
Gestation Weeks	26,290	31.78	2.19	20.00	42.00
$\geq$ 32 Gestation Weeks	26,290	0.53	0.50	0.00	1.00
Birth Weight in Grams	26,304	1507.53	76.91	1366.00	1634.00
Birth Weight < 1,500	26,304	0.44	0.50	0.00	1.00
Birth Length in cm	25,899	40.32	2.09	19.00	56.00
Death Within 1st Year of Birth	26,303	0.10	0.30	0.00	1.00
Days Spent in Hospital by Year 1	12,458	39.58	37.31	1.00	365.00
Number of Admissions to Hospital by Year 1	17,568	1.26	1.43	0.00	43.00
Mother's Age	26,304	28.02	7.13	15.00	45.00
Mother's Education Years	26,118	11.45	3.37	0.00	21.00
Panel B: Second Generation Births					
Gestation Weeks	1,625	38.38	2.11	22.00	42.00
$\geq$ 32 Gestation Weeks	1,625	0.98	0.13	0.00	1.00
Birth Weight in Grams	1,625	3185.99	553.66	490.00	4875.00
Birth Weight < 1,500	1,627	0.01	0.11	0.00	1.00
Birth Length in cm	1,625	48.68	2.89	20.00	56.00
Death Within 1st Year of Birth	1,627	0.01	0.09	0.00	1.00
Mother's Age	1,627	19.92	2.60	15.00	25.00
Mother's Education Years	1,625	11.22	2.21	0.00	17.00

Notes: Summary statistics are displayed for births occurring close to the 1,500 gram treatment threshold for the first generation (all births between 1992 and 2018), as well as those births matched to prior births (second generation births). The full sample consists of all births occurring between 1992 and 2018 in Chile from administrative data maintained by the Ministry of Health, and here we subset based on treatment assignment (mother's birth weight). This is based on the 134.4 and 227.5 gram optimal bandwidth cut-off when considering infant mortality for the first generation and Anderson index for the second generation. Hospitalisation days refer to days *only* for those births which are admitted to hospital.

Figure A2: Age of Mothers at Childbirth



(a) 1992



(b) 2018



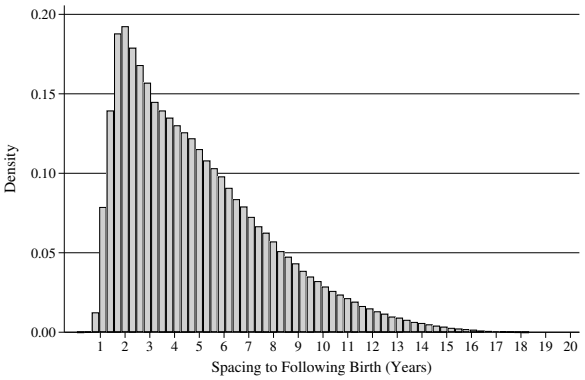
(c) 1992-2018



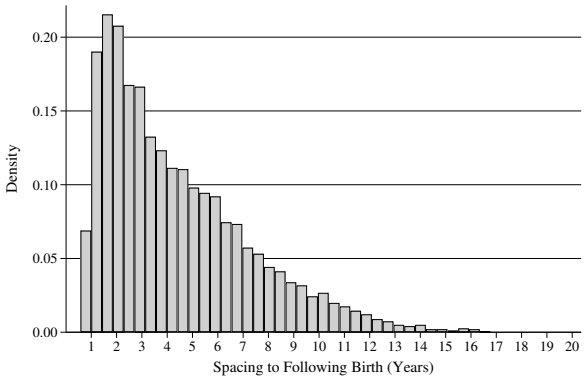
(d) Generation 1 Mothers

Notes: The distributions of births by mother's age are displayed. Births with mothers greater than age 25 are shown in green, while mothers aged 25 and below (corresponding to the maximum age in the intergenerational sample) are shown in red.

Figure A3: Observed Birth Spacing Between Children and their Future Siblings



(a) All Individuals



(b) $1300 \leq$ Birth weight ≤ 1700

Notes: Histograms document the space in years between each observed birth and following births of each mother (provided following births are observed) in the full sample of administrative records from birth registries (panel (a)), and only those births with babies weighing between 1,300 and 1,700 grams (panel (b)).

B Identification Checks

In this appendix we collect a range of tests relating to the validity of the very low birth weight RDD. Identification in a regression discontinuity design relies on the well-known continuity assumption related to unobservable factors around the cut-point (Hahn et al., 2001). If all unobservable factors are balanced on either side of the discontinuity, provided a small enough bandwidth is used for estimation, and provided that the cut-off generates a discontinuous jump in treatment, RD models can be used to isolate the impact of treatment on the dependent variable(s) of interest. In this appendix we first collect tests which document the relevance of the treatment in the first generation (Appendix B.1), and then a series of tests examining the identifying assumptions relating to treatment assignment to generation 1 mothers.

B.1 Proof of Treatment

Prior to considering any second generation outcomes, we seek to confirm previous evidence of the relevance of this treatment in this setting. This is a 'first-stage' which establishes the feasibility of later observing *intergenerational* spillovers of the initial treatment. Bharadwaj et al. (2013) have demonstrated that assignment to intensive neonatal care regimes due to crossing the VLBW threshold brought about sharp reductions in rates of infant mortality in Chile. We first document that these results can be replicated with the newly public matchable microdata files, on various samples of data, and using both their original and more recent optimal RD methods. We also document that these results are found among earlier generations, which is key in demonstrating the relevance of these treatments among cohorts who will be mothers in the intergenerational sample.

Figure B2 presents regression discontinuity plots which examine binned rates of infant mortality among births occurring in Chile around the 1,500 gram cut-off. Panel (a) replicate Bharadwaj et al. (2013)'s methods and definitions, working with the same sample of births between 1992-2007. Note that here Bharadwaj et al. (2013) bin weights in blocks of 30 grams, centred on points of 10 grams, with the exception of the point closest to the cut-off on either side, which is defined sharply. This implies that a single birth will be represented in multiple points. We follow their methods for comparability in panel (a), considering the same 100 gram range on either side of the cut-off, and in panels (b) and (c) use 10 mutually exclusive bins on either side of the treatment cut-off, and optimal bandwidth choices of Cattaneo et al. (2020). In all cases, a clear increase in infant mortality is observed when crossing the 1,500 gram threshold.

For our purposes, what is key is ensuring the relevance of initial treatment, which is clear even graphically in Figure B2. Formal RD estimates are presented in Appendix Table B1. Estimates are presented based on all three samples presented in Figure B2, estimating that, in the 'intergenerational sample', infant mortality falls by 2.7 pp, compared to a base of 13.8pp, or by approximately 22%, clearly establishing the relevance of these treatments in this particular setting. If we alternatively examine only earlier years, for example 1992–2001, similar results are observed. See Figure B3. If extending these tests to consider rates of infant death up to 2018, the results first documented in Bharadwaj et al. (2013) are, if anything, strengthened, with column 3 of Table B1 estimating a 2.7 pp reduction in rates of infant death against a base of 10.2 pp, even within the very tight bandwidth of 100 grams used by Bharadwaj et al. (2013), with very similar results observed in column 4 when estimating based on the optimal bandwidth of 134.4 grams. Note that these effect sizes are large, of the order of magnitude of a 20-30% decline in rates of infant death. A simple back of the envelope cost-benefit calculation presented in Online Appendix F suggests that these benefits from treatment receipt in the first generation dwarf any future costs incurred as a result of the policy given required investments in future generations. This holds even if we additionally allow for negative spillovers from fathers to their children.

Estimation Sample As we noted in section 2, and as laid out in Bharadwaj et al. (2013), there is clear evidence of a sharp cut-off in treatment rules at 1,500 grams when babies are born at 32 weeks or above. In certain cases, babies born at below 32 weeks will access the same treatment regimes, whether they are born at above or below the 1,500 gram threshold. In main estimates we thus focus exclusively on the sample of individuals born at 32 weeks and above, and in which case the discontinuity certainly binds. This sample covers 56% of all births within 200 grams of the 1,500 gram threshold. However, there is some evidence to suggest that at least in a more limited way, there is a discontinuity in treatment around 1,500 grams for babies born at below 32 weeks. We document this in Figure B4 which shows rates of hospitalisation by age, with evidence that—in line with policy definitions—babies born at less than 1,500 grams have elevated rates of hospitalisation early in life, and in particular in the first year of life, consistent with more intensive treatment. Thus, as a robustness check, we additionally present results from the main text covering the full sample of individuals, regardless of their gestational length.

B.2 Identification Checks

Identifying the impacts of marginal medical investments at birth on subsequent outcomes based on our RD design requires that no other factors vary sharply local to the 1,500 gram cut-off. We consider a number of tests of these identifying assumptions, though note that this design has previously been validated in this and other settings by Bharadwaj et al. (2013); Chyn et al. (2021); Almond et al. (2010); Daysal et al. (2022). In particular, our tests here focus on two considerations: firstly, is there balance at treatment receipt of observable characteristics, and secondly, is there evidence of manipulation of birth weights suggesting that there may be systematic, or strategic, selection into treatment by medical practitioners, or by families of babies born in this bandwidth.

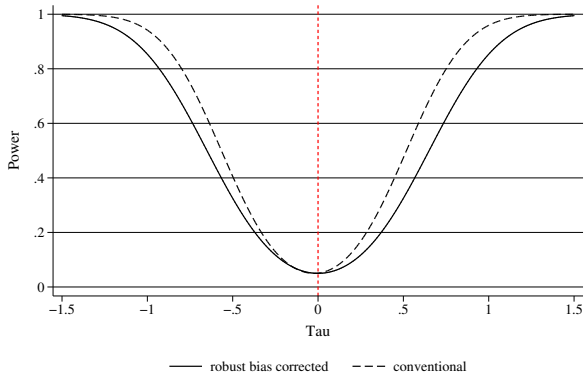
A first consideration which is key is examining the predetermined characteristics of individuals who are located just below the 1,500 gram threshold, and hence receive of intensive medical intervention, and those located just above the threshold. We would be concerned if we observed that certain individuals are more likely to receive access to the policy, as it may illustrate imperfect compliance with the treatment threshold, and confound estimates presented throughout this paper. We examine a number of measures of first generation mothers, fathers and births, presented graphically in Appendix Figure B5, with associated RD tests in Appendix Table B2. Specifically, we do not see that treated individuals have parents of substantially different ages or educational levels or who are more closely aligned in age, nor do we see that they are more likely to be married, or are more likely to be born in urban areas. This balance point is also highlighted by Bharadwaj et al. (2013).

Nevertheless, a general concern in this case is still related to manipulation of the running variable, potentially not captured by these observable factors. If parents or medical practitioners were able to manipulate the running variable (officially recorded birth weight), they may be able to selectively ensure coverage for certain types of individuals. If this is the case, and if such manipulation owed to visual clues or information from patient histories inferred by medical practitioners, one may suspect that babies registered as having weights just to the left of the cut-off look different in birth size or gestational length than those just to the right. In Figure B6 we observe no evidence to suggest that this is the case. We also note that in the case of manipulation, one would expect a greater likelihood of observing births at points close below 1,500 grams, and a lower likelihood of observing births just above 1,500 grams, which is not something we observe. As proposed by Almond et al. (2011), we conduct tests formally examining whether there are more births observed in micro-data registered as having birth weight just to the left compared to just to the right of the cut-off, and find no evidence to suggest that this is the case (Appendix Table B3).

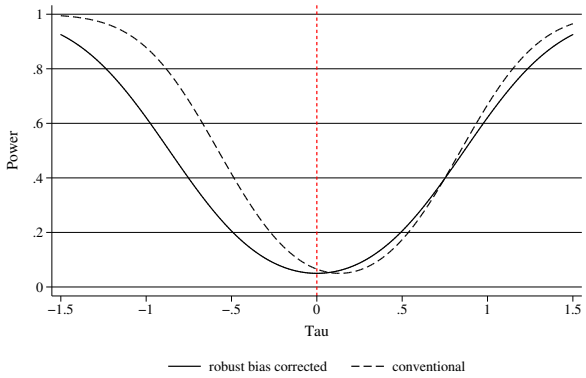
Another specific concern related to manipulation and measurement in this setting, discussed extensively in the existing literature, is the presence of heaping in birth weight (Barreca et al., 2011, 2016; Almond

et al., 2011). This can be observed when examining simple plots of the frequency of individuals observed at particular birth weights. In Appendix Figure B7 we observe that while birth weight is regularly distributed when zooming out across the entire distribution, regular peaks are observed at 50 and 100 gram bins, apparent when zooming in on birth weight in panel (b). Similarly, Bharadwaj et al. (2013) note that this rounding occurs differentially in certain types of hospitals, implying that it may be not be innocuous in estimation. We note that in the case of assignment rules, individuals will receive treatment only if they are observed to have a birth weight just *below* 1,500 grams. Given this, throughout this paper we control flexibly for heaping and include controls for demographic factors proposed by Bharadwaj et al. (2013). But to ensure that our estimates are not driven by heaping, specifically that at 1,500 grams, we estimate the 'Donut' RD models suggested by Barreca et al. (2011, 2016), removing observations which are located very close to the cut-off. We discuss these models when discussing robustness of our estimates in Section 5.1.2.

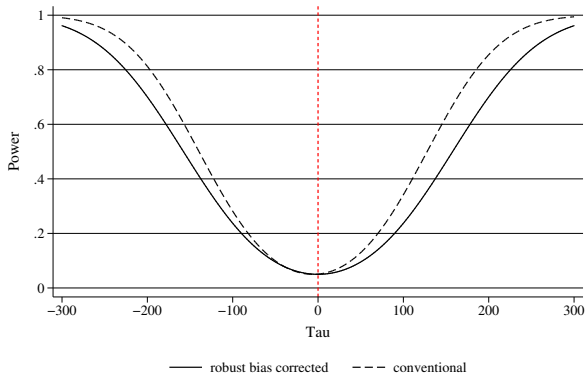
Figure B1: Power Analysis – RD Models of Intergenerational Impacts



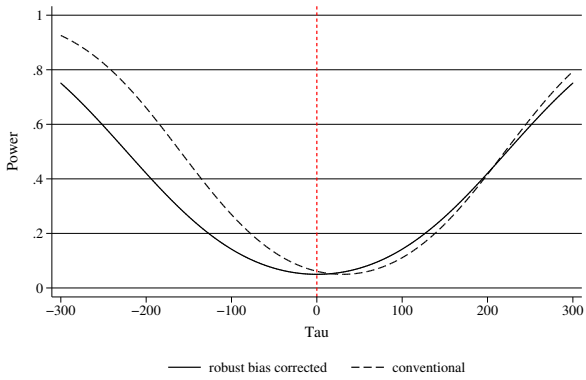
(a) Child's Gestational Length, Baseline



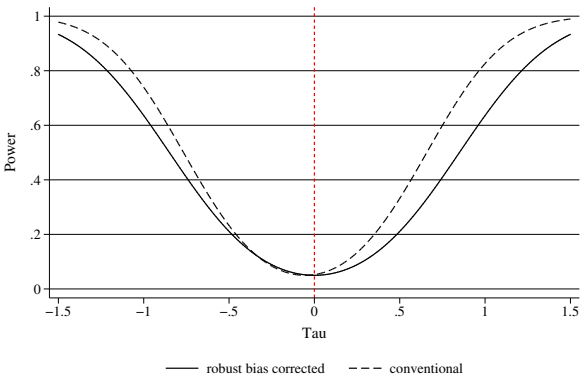
(b) Child's Gestational Length, Heaping Control



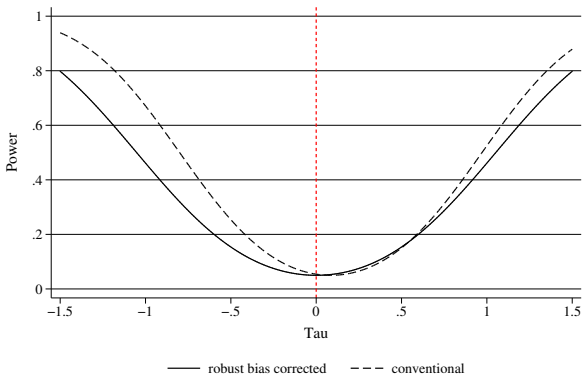
(c) Child's Birth weight, Baseline



(d) Child's Birth weight, Heaping Control



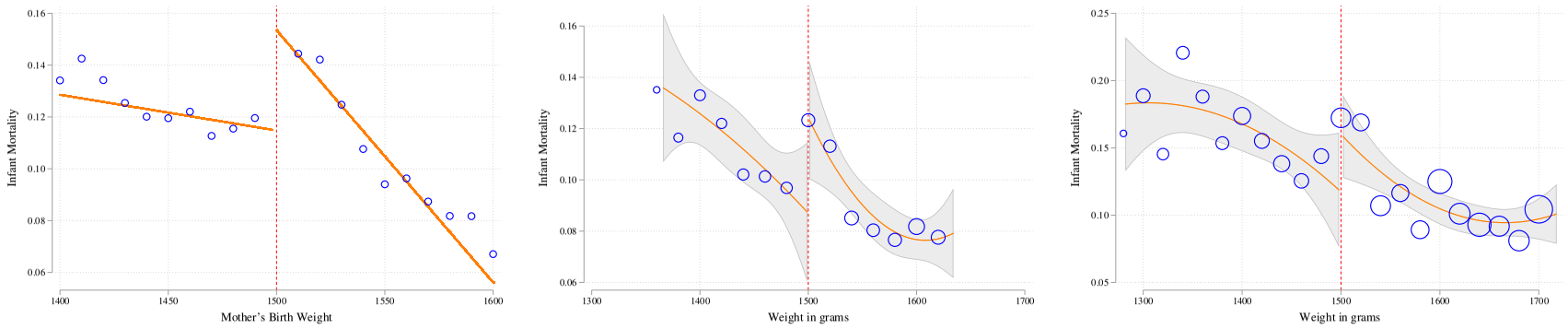
(e) Child's Birth length, Baseline



(f) Child's Birth length, Heaping Control

Notes: Curves in each figure plot the power of our regression discontinuity design to detect effect sizes indicated by Tau on the horizontal axes against a null of zero effects. Effects refer to intergenerational impacts, and as such, in each case, samples consist of second generation births corresponding to our principal estimation sample. Power calculations follow Cattaneo et al. (2019b).

Figure B2: Birth weight Assignment Thresholds and Infant Mortality



(a) BLN Method and Sample

(b) Full Period and Optimal Bandwidth

(c) Early Cohorts and Optimal Bandwidth

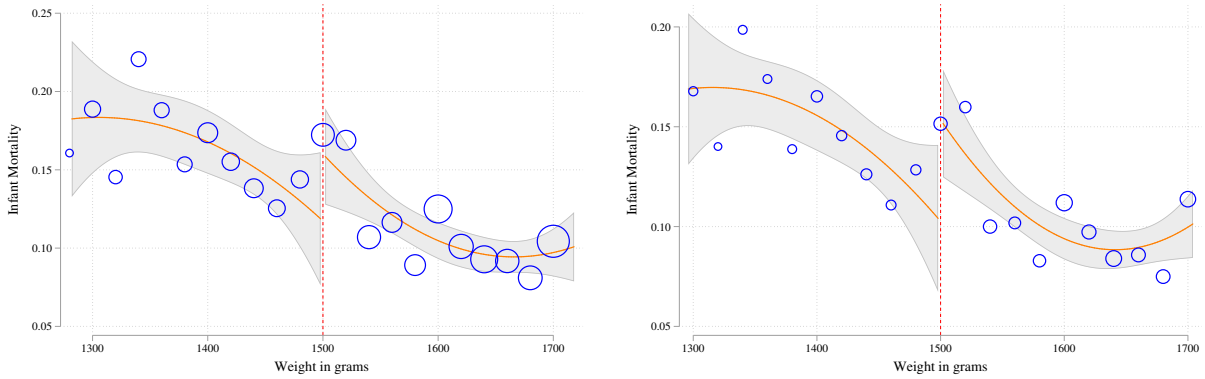
Notes: Each sub-plot examines the impact of crossing the VLBW threshold on infant mortality. All panels present estimates for gestational weeks 32 and above (where assignment rules clearly apply). Panel (a) replicates Bharadwaj et al. (2013)'s methods using overlapping (30 g) bins and a 100 gram bandwidth. Panel (b) uses optimal bandwidth selection methods of (Calonico et al., 2020), and plots a quadratic fit with 95% CIs, where point sizes are indicative of relative sample sizes. Panel (c) replicates the optimal plot from panel (b) but focusing only on earlier birth cohorts, who are represented as mothers in the intergenerational sample.

Table B1: Initial Policy Impacts on Infant Mortality Rates

	Bharadwaj et al. (2013) Period		Full Period (1992-2018)		Early cohorts (1992-2001)	
	(1)	(2)	(3)	(4)	(5)	(6)
Conventional	-0.0311**	-0.0165*	-0.0270***	-0.0207***	-0.0266*	-0.0176
	(0.0123)	(0.0099)	(0.0080)	(0.0075)	(0.0152)	(0.0109)
Robust	-0.0131	-0.0189	-0.0295**	-0.0248***	0.0031	-0.0198
	(0.0147)	(0.0117)	(0.0120)	(0.0085)	(0.0219)	(0.0128)
Mean of Dep. Var.	0.120	0.111	0.102	0.098	0.138	0.127
Observations	3,913,016	3,913,016	6,525,049	6,525,049	2,554,926	2,554,926
Optimal Bandwidth	100.0	203.6	100.0	134.4	100.0	219.3
Effective Observations	5,340	12,843	9,464	13,815	3,431	8,809
Observations (left)	2,037	3,942	3,695	4,999	1,279	2,646
Observations (right)	3,303	8,901	5,769	8,816	2,152	6,163
Bharadwaj et al. (2013) Procedure	Y		Y		Y	
Optimal Bandwidth RBC		Y		Y		Y

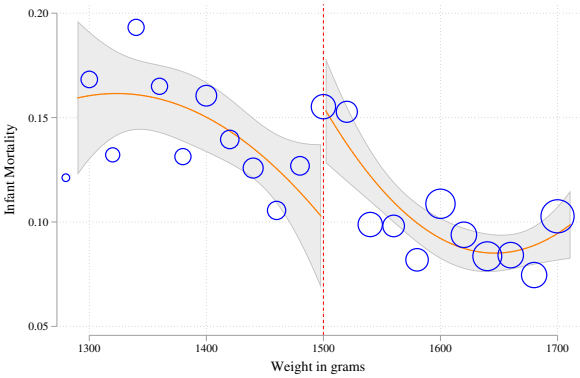
Notes: Each column presents estimates of the impact of intensive treatment owing to the VLBW assignment on an indicator for infant mortality. Models in odd columns replicate the methods of Bharadwaj et al. (2013) using a 100 gram cut-off, and in even columns use the MSE optimal bandwidth of Calonico et al. (2020). In each case standard RD models are presented (top rows) which correspond to methods implemented by Bharadwaj et al. (2013), and robust bias corrected estimates and standard errors are presented below. Columns (1) and (2) use the exact same time period as that studied by Bharadwaj et al. (2013) (1992-2007), columns (3) and (4) consider the full sample of data used in this paper, and columns (5) and (6) consider only birth cohorts in which second generation mothers are born. Significance stars for two-sided test: * $p<0.10$; ** $p<0.05$; *** $p<0.01$.

Figure B3: Birth weight Assignment Thresholds and Infant Mortality (Early Years Only)

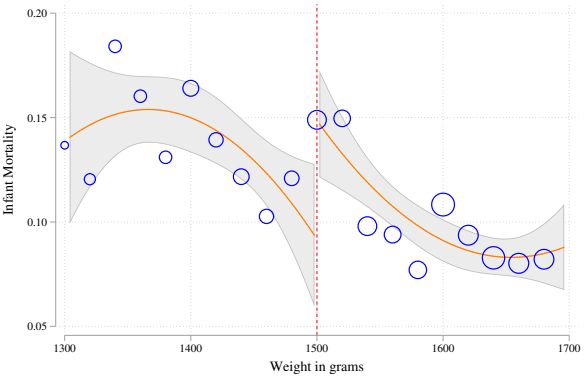


(a) $\leq$ 2001 and Optimal Bandwidth

(b) $\leq$ 2004 and Optimal Bandwidth



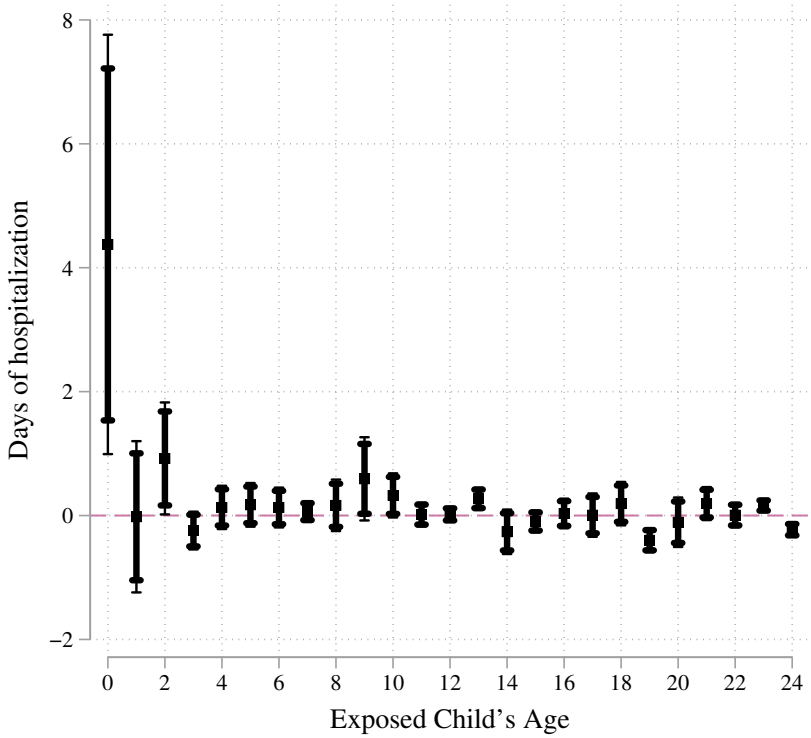
(c) $\leq$ 2005 and Optimal Bandwidth



(d) $\leq$ 2006 and Optimal Bandwidth

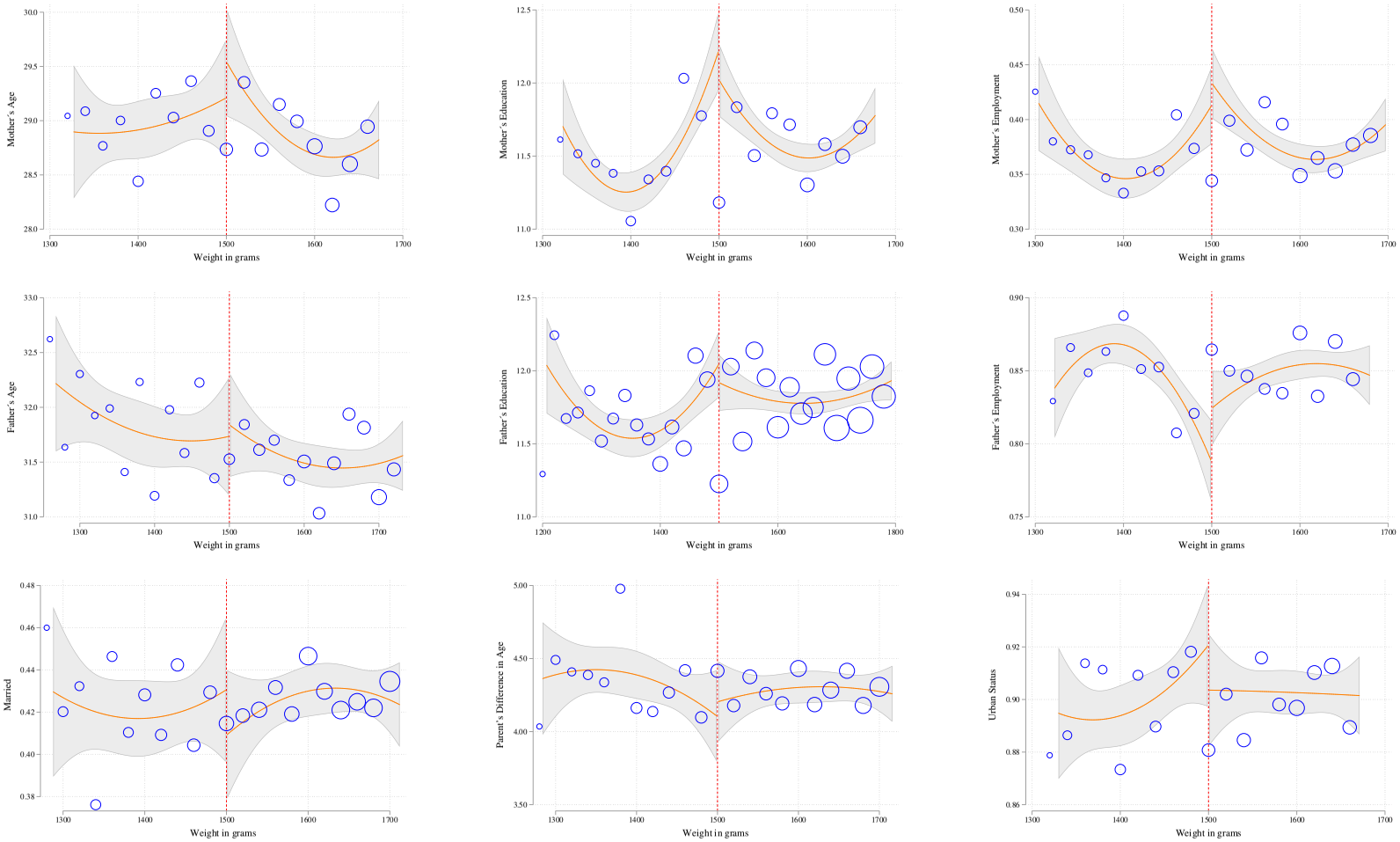
Notes: Refer to Notes to Figure B2. Identical plots are presented to panel (c), however here documenting alternative considerations of "early cohorts", as indicated in captions.

Figure B4: Discontinuities in Hospitalisation in Babies Born at $<$ 32 Gestational Weeks



Notes: Each point estimate and confidence interval refers to the impacts of early life medical investment on an individual's days of hospitalisation at each age, estimated by RD using the full sample of births *born at less than 32 gestational weeks* using the RD specification. Thicker black error bars present 90% CIs, while thinner error bars report 95% CIs. Days of hospitalisation are measured as totals for all individuals who have reached the age indicated, and take the value of 0 if the individual is not hospitalised this year, or otherwise a positive integer reporting the total number of days spent in hospital. All estimates follow the procedures laid out in section 4, and report RBC estimates using a local linear regression with a triangular kernel in the MSE optimal bandwidth.

Figure B5: Balance Tests – Generation 1 Family and Birth Characteristics



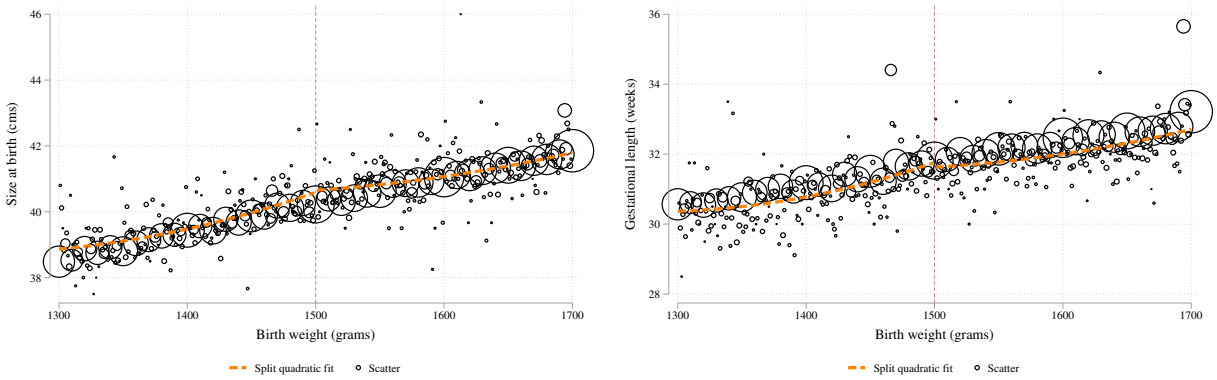
Notes: Plots document balance tests examining characteristics of individuals within the Calonico et al. (2020) RBC optimal bandwidth of the 1,500 gram treatment threshold. Binned averages for each outcome are presented in 20 gram birth weight bins, with circle sizes documenting the relative proportion of individuals in each bin. A separate split quadratic and confidence intervals are documented on each side of the cut-off. Formal tests of discontinuities are presented in Table B2.

Table B2: Balance Tests – Generation 1 Family and Birth Characteristics

	Mother			Father				Parent's	Observed	Urban
	Age	Education	Employed	Age	Education	Employed	Married	Age Diff.	Dad	Status
Birth weight < 1,500	0.135	0.659*	0.018	-0.012	0.553	-0.066**	0.007	-0.227	-0.030**	0.020
	(0.311)	(0.362)	(0.035)	(0.337)	(0.376)	(0.026)	(0.018)	(0.183)	(0.012)	(0.016)
Mean of Dep. Var.	28.836	11.537	0.371	31.542	11.728	0.851	0.424	4.308	0.948	0.895
Observations	6,084,923	6,077,898	6,084,923	5,816,066	5,451,503	6,084,923	6,084,923	5,816,066	6,084,923	6,084,923
Optimal Bandwidth	262.6	175.4	239.7	326.0	207.3	156.2	265.7	205.0	139.1	311.5
Effective Observations	28,010	17,062	24,360	34,334	18,425	15,335	28,149	19,892	13,149	34,783
Observations (left)	7,534	5,688	7,082	7,842	5,656	5,312	7,558	6,118	4,749	8,170
Observations (right)	20,476	11,374	17,278	26,492	12,769	10,023	20,591	13,774	8,400	26,613
q-sharpened p-value	0.632	0.225	0.632	0.635	0.330	0.074	0.632	0.372	0.074	0.372

Notes: Each column displays estimates of the change in the given dependent variable from above to below the 1,500 gram assignment threshold for newborns. Each outcome refers to mother, father, or family level characteristics of all first generation observations, and as such are balance tests examining assignment to treatment. In each case, local linear regression is used with a triangular kernel, calculating the MSE optimal bandwidth of Calonico et al. (2020). Robust bias corrected standard errors clustered at the gram level are reported in parentheses. Significance stars for two-sided test: * $p<0.10$; ** $p<0.05$; *** $p<0.01$.

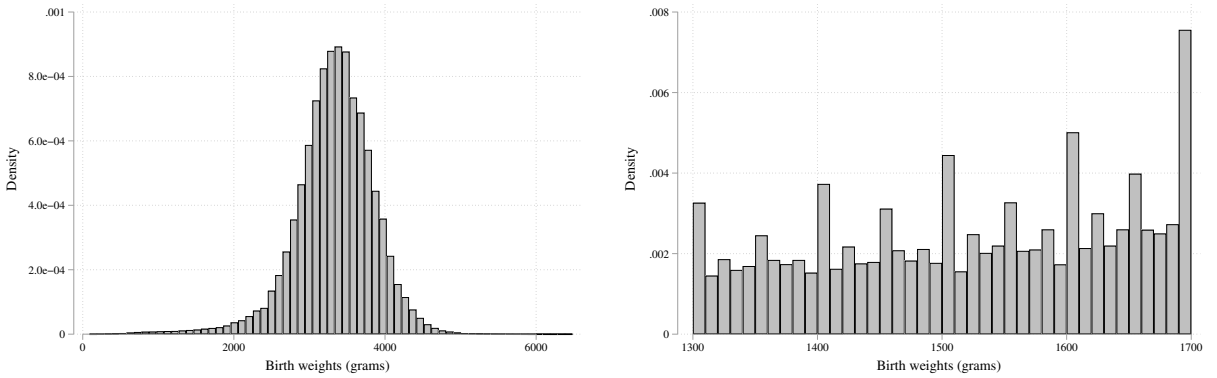
Figure B6: Observable Birth Outcomes by Birth Weight



(a) Birth size (cms) **(b)** Gestational weeks

Notes: Scatter plots are presented where each point represents average birth size (panel (a)), or gestational weeks (panel (b)) in 50 gram birth weight bins. Each panel is based on the full sample of births from 1992–2018 between 1,300-1,700 grams, and point sizes reflect the relative frequency of the sample in each bin. Separate linear trends are plotted on each side of the 1,500 gram VLBW threshold.

Figure B7: Birth Weight Frequency in Administrative Records



(a) Full Sample **(b)** 1300-1700 grams

Notes: Density plots are presented based on the full sample of births from 1992–2018. Panel (a) includes all births, while panel (b) limits only to births recorded as weighing between 1300 and 1700 grams (inclusive). In panel (b) 10 gram bins are plotted in the histogram.

Table B3: Testing for birth heaping at the cut-off

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Total Difference										
$<$ 1,500 gram threshold	9.400	2.900	-0.433	-0.275	-2.320	-5.613	-11.330	-14.368	-18.860	-27.230
	(81.499)	(64.159)	(51.153)	(44.892)	(44.899)	(34.172)	(33.774)	(29.073)	(27.374)	(24.469)
Total Births	1,610	3,498	5,251	7,219	9,638	13,401	19,703	23,634	29,415	39,934
Gram-Specific cells	20	40	60	80	100	150	200	250	300	400
R-Squared	0.001	0.000	0.000	0.000	0.000	0.000	0.001	0.001	0.002	0.003
Panel B: Total Difference and Distance Trend										
$<$ 1,500 gram threshold	-3.667	12.863	13.529	9.062	9.102	6.694	10.915	7.935	9.041	10.961
	(160.284)	(131.242)	(104.936)	(91.602)	(91.013)	(69.433)	(68.121)	(58.670)	(55.099)	(49.135)
Absolute Difference (grams)	29.673	10.253	4.586	2.841	2.580	0.565	0.936	0.416	0.472	0.401
	(18.266)	(7.747)	(4.180)	(2.753)	(2.196)	(1.123)	(0.828)	(0.571)	(0.448)	(0.300)
Absolute Difference (grams) $\times$ $<$ 1,500 grams	2.376	-0.949	-0.901	-0.455	-0.448	-0.324	-0.440	-0.354	-0.370	-0.380
	(25.832)	(10.956)	(5.911)	(3.894)	(3.106)	(1.588)	(1.171)	(0.808)	(0.633)	(0.424)
Total Births	1,610	3,498	5,251	7,219	9,638	13,401	19,703	23,634	29,415	39,934
Gram-Specific cells	20	40	60	80	100	150	200	250	300	400
R-Squared	0.264	0.082	0.034	0.023	0.024	0.002	0.009	0.003	0.006	0.008
Absolute margin (grams)	10	20	30	40	50	75	100	125	150	200

Notes: Each column presents a regression of the number of births observed on an indicator for whether the recorded weight is below 1,500 grams, in a fixed range around the 1,500 gram cut-off. This range is varied across columns from$\pm$ 20 grams to $\pm$ 400 grams, as indicated in the table footer. In panel (A) a single binary "$<$ 1,500 gram" indicator is included, while in panel (B) this is additionally interacted with a control for absolute difference in grams, to control for the fact that the number of births gradually increase as birth weight increases. Significance stars for two-sided test: * p$<$0.10; ** p$<$0.05; *** p$<$0.01.

C Additional Model Details

C.1 A Single Generation Model

Previous literature in this setting (Bharadwaj et al., 2013; Chyn et al., 2021) has proposed a first generation model consisting of initial endowments, medical treatments (which interact with arbitrary treatment assignment cut-offs), and subsequent post-hospital investments. To understand the impacts of treatment receipt on a treated individual during their own life, consider the stylised setting where the outcome of interest is an individual's (own) health. In a simple framework, along the lines of Grossman (2000), suppose that these time-varying health measures denoted H_{it} depend on initial endowments (H_i), medical intervention at birth (D_i), as well as subsequent parental investment.[21] As Bharadwaj et al. (2013) note, parental investment can interact with initial medical treatments in such a way to reinforce or compensate initial treatment receipt. Following their notation, parental investment is denoted $I_t^{post}(H, D, \zeta)$, capturing accumulated investments up to t which may be a function of initial health, treatment at birth, and subsequent shocks ζ. Thus, 'first generation' health is modelled as:

$$H_{it} = \phi_t H_i + \psi_t D_i + \varphi_t I_t^{post}(H_i, D_i, \zeta_i) + X_{it}\beta_t + \upsilon_{it},$$

where it is clear that individual health at time t, H_{it} may depend on initial treatment directly, but also in the way that this treatment D_i interacts with subsequent parental behaviour.

As in Bharadwaj et al. (2013), while the RD isolates an exogenous shift in treatment intensity, the coefficient of interest from (1) when considering H_{it} as the outcome of interest captures the full policy impact up until t:

$$\widehat{\alpha} = \psi_t \cdot \kappa + \varphi_t \cdot \Delta I_t^{post}(c),$$

consisting of both structural effects from treatment at birth, as well as differences in posterior parental investments surrounding the treatment cut-off.

Thus, when considering health outcomes of treated women after birth but prior to any second generation births, the estimated coefficient $\widehat{\alpha}$ will be interpreted as the total policy relevant treatment effect, based on individual and parental response up until the moment that the outcome of interest is measured.

C.2 Greater Granularity in Fertility Choices

In Section 4.2 we develop a model in which selection is allowed to differ when considering fertility at younger and at older ages. This definition can be further generalised without any major impacts on the modelling implications insofar as they drive our consideration of potential mechanisms. To see this, note that instead of (2)-(3) we could write age-specific fertility as a function of individual's health, early life investments, parental investments and other behaviours as:

$$\text{Fertility at Age X}_i^* = \gamma_1^X H_i + \gamma_2^X D_i + \gamma_3^X I^{post}(H_i, D_i) + \gamma_4^X B^{post}(H_i, D_i) + \upsilon^X. \tag{8}$$

for all X observed. A key point here is that this permits each of these determinants to affect fertility in a different way throughout an individual's life. In this case, if birth weight is modelled as in (4), this will lead to a separate expected health outcome at birth for each age X, written as:

$$\begin{aligned} E[BW_{ij}|\text{Fertility at Age X}_i = 1] &= H_{ij} + \phi H_i + \psi D_i + \varphi I^{post}(\cdot) + \Psi B^{post}(\cdot) \\ &\quad + \rho_X \sigma \lambda_X \left[H_i, D_i, I^{post}(\cdot), B^{post}(\cdot)\right], \end{aligned}$$

[21] In Grossman's canonical health capital model, health at time $t+1$ depends upon health at time t as well as (recursively) health at birth. Here given we are interested in estimating impacts of early life interventions we focus principally on health at birth. However, as we show below, we can incorporate dynamic health flows in this model without greatly altering the implications of this model for understanding RDD estimates.

and now, selection into fertility may vary *at each age* X via distinct selection terms $\lambda_X(\cdot)$. The implications of this would be identical to those discussed at the end of Section 4.2, simply with a richer channel to describe birth timing.

C.3 Dynamic Health Stocks

Here we note how the model laid out in section 4 of the paper generalises to a more flexible Grossman (2000)-style health model, where individual health is measured as a dynamic flow, depending on health at birth, as well as health in previous periods. Consider health at adulthood, when women are at risk of falling pregnant, and when one wishes to consider the interpretation of the RD model laid out in the paper, examining the intergenerational transmission of health at birth. In this case, the analogue of health stocks of generation 1 laid out in Appendix C.1 is:

$$H_{it} = \phi_t H_i + \mu_t H_{i,t-1}(H_i, D_i, I_t^{post}, \zeta_i) + \psi_t D_i + \varphi_t I_t^{post}(H_i, D_i, H_{i,t-1}, \zeta_i) + X_{it}\beta_t + \upsilon_{it}, \tag{9}$$

where now, H_{it} is explicitly a function of $H_{i,t-1}$, and in turn, depends recursively on health in all previous periods: $H_{it} \equiv H_{it}(H_{i,t-1}, H_i, D_i, I_t^{post}, X)$. We note two specific relevant considerations of this model of health stocks. Firstly, it makes explicit that like parental investments, previous health stocks may depend upon initial health, treatment at birth, and subsequent stocks, as well as parental investment. Importantly, this implies that H_{it-1} may shift owing to threshold crossing, which must be taken into account when considering the interpretation of a reduced form RDD parameter during adulthood. Secondly, we note that health stocks at birth are explicitly indicated as H_i, to note the lack of dependency on these response variables.

Now, as laid out in the paper, equations 2-4 capture the implications of selection into fertility on the estimated intergenerational impacts of health investment at birth. To see how this impacts the final decomposition of the estimated RDD parameter $\widehat{\alpha}$, we generalise the model here to incorporate $H_{i,t-1}$. First, note that latent fertility now depends upon accumulated health stocks:

$$\text{Early Fertility}_i^* = \gamma_1^E H_i + \gamma_2^E D_i + \gamma_{3t}^E H_{i,t-1}(H_i, D_i, I_t^{post}, \zeta_i) + \gamma_{4t}^E I_t^{post}(H_i, D_i, H_{i,t-1}, \zeta_i) + X_{it}\pi_t^E + \iota_{it}^E, \tag{10}$$

with a similar equation for Late Fertility.[22] Similarly, the model of birth weight of generation 2 also depends upon the mother's accumulated health stocks:

$$BW_{ij} = H_{ij} + \phi_t H_i + \mu_t H_{i,t-1}(H_i, D_i, I_t^{post}, \zeta_i) + \psi_t D_i + \varphi_t I_t^{post}(H_i, D_i, H_{i,t-1}, \zeta_i) + X_{it}\beta_t + u_{ij}, \tag{11}$$

with BW_{ij} observed for individuals only if Early Fertility$_i = 1$. Following the model in the paper, the conditional expectation of birth weight is now:

$$\begin{aligned} E[BW_{ij}|\text{Early Fertility}_i = 1] \quad = \quad & H_{ij} + \mu_t H_{i,t-1}(H_i, D_i, I_t^{post}, \zeta_i) + \phi_t H_i + \\ & \psi_t D_i + \varphi_t I_t^{post}(H_i, D_i, H_{i,t-1}, \zeta_i) + X_{it}\beta_t + \\ & \rho_E \sigma \lambda_E[H_i, H_{i,t-1}(H_i, D_i, I_t^{post}, \zeta_i), D_i, I^{post}(H_i, D_i, \zeta_i)] \end{aligned} \tag{12}$$

where all notation follows section 4.2 of the paper. Finally, in this case, the extended interpretation of the RDD estimate $\widehat{\alpha}$ is now the following:

$$\widehat{\alpha} = \psi_t \cdot \kappa + \mu_t \cdot \Delta H_{it-1}(c) + \varphi_t \cdot \Delta I_t^{post}(c) + \rho_E \sigma \Delta\lambda_E[c, \Delta H_{it-1}(c), \Delta I_t^{post}(c)] + \rho_L \sigma \Delta\lambda_L[c, \Delta H_{it-1}(c), \Delta I_t^{post}(c)]. \tag{13}$$

There are two upshots from this extended model allowing for dynamics in mother's health. Firstly, this opens up a direct new channel which may partially explain the RD estimate, which is that mother's health at $t-1$

[22] For simplicity we are omitting the dependence on $B^{post}()$ in this section, though its inclusion would not affect the results here.

(and recursively before that) impacts child's health at t, given that health investments at birth may directly impact mother's health at time t. This is indicated by $\mu_t \cdot \Delta H_{it-1}(c)$. Secondly, it additionally opens up a new sub-channel within the inverse Mills ratio, which is that a mother's health stock at time $t-1$ makes her more or less likely to give birth. This is indicated by $\Delta H_{i,t-1}(c)$ within the inverse Mills ratio, where this is relevant if treatment receipt shifts later-life health around the cut-off c. Note then that we can 'close off' this channel if we can show that women are not more or less healthy at each time t given treatment receipt.

C.4 Proceeding without a Normality Assumption

In the main model laid out in the paper, selection into fertility was accounted for using the well-known Heckman (1974) correction, which relies on the assumption that the unobservables in the selection equations and the outcome equation are jointly normal. Under this assumption, selection enters the outcome equation through the inverse Mills ratio, which is a function of the estimated selection residuals.

However, such an assumption is simply taken as it provides a convenient form for the selection equation. It is, however, not required for the theoretical implications discussed at the end of Section 4.2. To generalise this framework and relax the normality assumption, we can use a control function approach. Rather than relying on the inverse Mills ratio, we introduce a flexible correction term that accounts for selection in a more general way. Specifically, we replace the inverse Mills ratio with a nonparametric control function that depends on the estimated selection residuals. The modified expectations of second-generation birth weight conditional on selection are given by:

$$\begin{aligned} E[BW_{ij}|\text{Early fertility}_i = 1] &= H_{ij} + \phi H_i + \psi D_i + \varphi I^{post}(\cdot) + \Psi B^{post}(\cdot) \\ &\quad + CF_E\left(H_i, D_i, I^{post}(\cdot), B^{post}(\cdot), \widehat{v}^E\right), \end{aligned} \tag{14}$$

$$\begin{aligned} E[BW_{ij}|\text{Later fertility}_i = 1] &= H_{ij} + \phi H_i + \psi D_i + \varphi I^{post}(\cdot) + \Psi B^{post}(\cdot) \\ &\quad + CF_L\left(H_i, D_i, I^{post}(\cdot), B^{post}(\cdot), \widehat{v}^L\right). \end{aligned} \tag{15}$$

Here, $CF_E(\cdot)$ and $CF_L(\cdot)$ are unspecified control functions that capture the selection effects without imposing a specific functional form. These functions depend on the estimated residuals $\widehat{v}^E$ and $\widehat{v}^L$ from the first-stage selection equations. Unlike the inverse Mills ratio, which arises from the normality assumption, these control functions allow for a more flexible correction by incorporating potential nonlinearities or interactions. This also would allow for IV-based procedures to be used in conjunction with control functions.

While this change the way that we write (5)-(6), it does not alter the implications given that now our estimated effect from Section 4.2 would be written as follows:

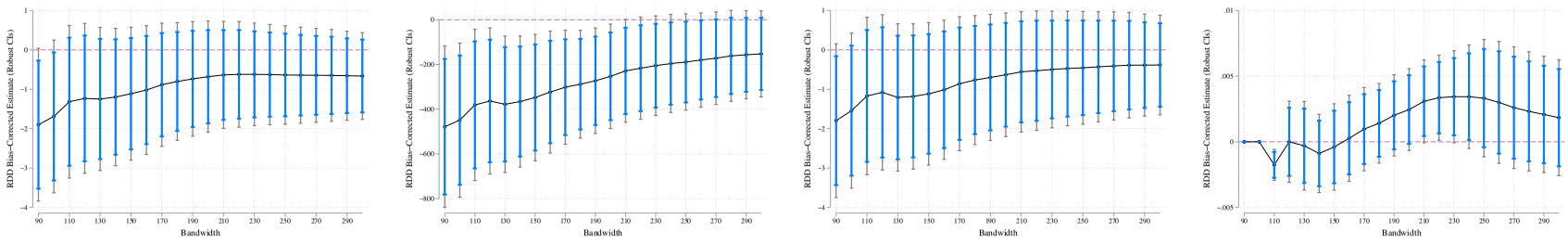
$$\begin{aligned} \widehat{\alpha} \quad = \quad & \psi\kappa && \text{"Structural"} \\ + \quad & \varphi \cdot \Delta I^{post}(c) + \Psi \cdot \Delta B^{post}(c) && \text{"Behavioural"} \\ + \quad & \Delta CF_E\left[c, \Delta I^{post}(c), \Delta B^{post}(c), \Delta\widehat{v}^E(c)\right] && \text{"Selection (Early Fertility)"} \\ + \quad & \Delta CF_L\left[c, \Delta I^{post}(c), \Delta B^{post}(c), \Delta\widehat{v}^L(c)\right] && \text{"Selection (Later Fertility)"} \end{aligned}$$

Here, as in the model laid out in the paper, the key point is that any selection into fertility is relevant to the treatment effect if this selection is related to the outcome of interest, and any differential selection by age is similarly relevant if this is related to health outcomes at birth.

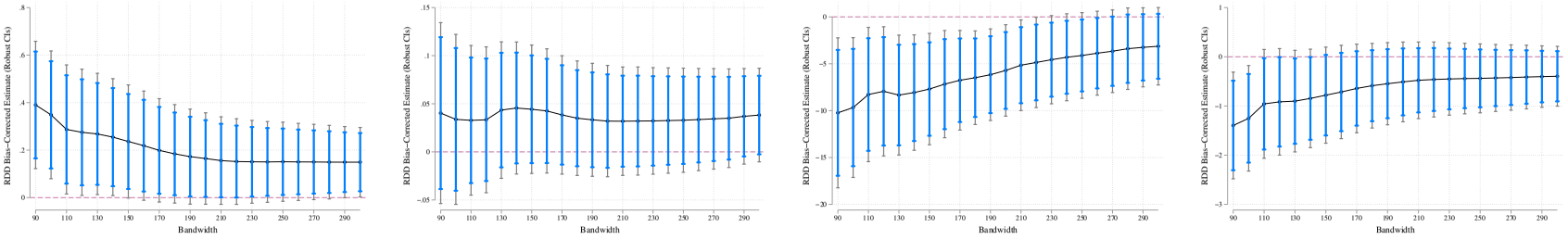
In the body of the paper, our interest in this model is as a way to consider potential channels discussed in the mechanism Section 5.2. We thus do not estimate the control function approach (or the Heckman selection approach laid out in the text), but rather evaluate whether evidence exists to support each of the channels posited by the model. We nevertheless discuss the implications of Selection into treatment for the estimated parameter $\widehat{\alpha}$ in Section 5.3 of the text.

D Robustness Checks

Figure D1: RD Estimates of Intergenerational Impacts Varying Estimation Bandwidths



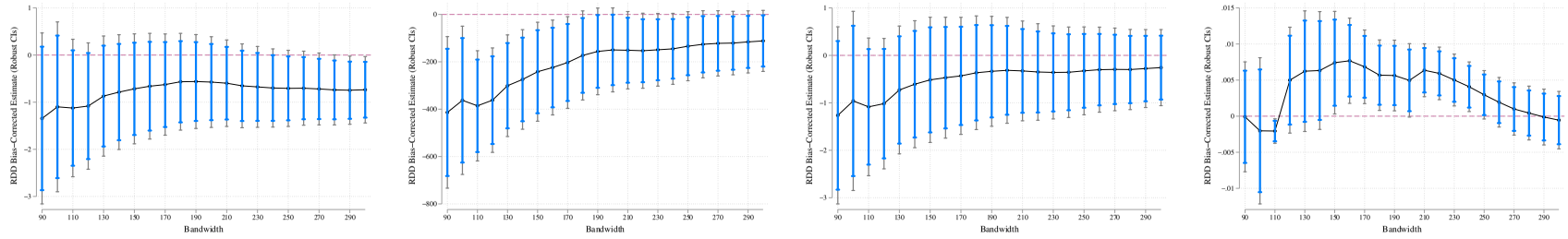
(a) Gestation (weeks) **(b)** Birth weight (grams) **(c)** Birth length (cms.) **(d)** Infant mortality



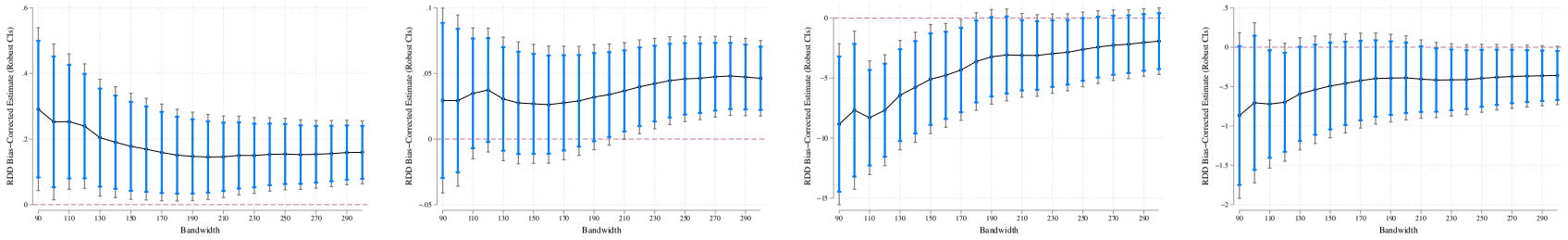
(e) Prematurity **(f)** Very low birth weight **(g)** Fetal growth rate **(h)** Anderson index

Notes: Each sub-plot portrays the impact of crossing the VLBW threshold on a specific second-generation health measure from Table 2, where the RDD specification is estimated within a bandwidth manually set to 90,100,110,…,300 grams. Point estimates and 90% and 95% confidence intervals (blue and grey lines respectively) are shown using bias corrected estimates with heaping controls. The estimation sample consists of all children of first generation mothers who were born at 32 weeks or greater of gestation.

Figure D2: RD Estimates of Intergenerational Impacts Varying Estimation Bandwidths with Constant Relative Bias



(a) Gestation (weeks) **(b)** Birth weight (grams) **(c)** Birth length (cms.) **(d)** Infant mortality



(e) Prematurity **(f)** Very low birth weight **(g)** Fetal growth rate **(h)** Anderson index

Notes: Each sub-plot portrays the impact of crossing the VLBW threshold on a specific second-generation health measure from Table 2, where the RDD specification is estimated within a bandwidth manually set to 90,100,110,…,300 grams. Point estimates and 90% and 95% confidence intervals (blue and grey lines respectively) are shown using bias corrected estimates with heaping controls, where the bias correction calculation is conducted by holding a constant relative bias correction range to optimal bandwidth range as that used in principal models displayed in the paper. The estimation sample consists of all children of first generation mothers who were born at 32 weeks or greater of gestation.

Table D1: Alternative Specifications – Gestational weeks, birth weight, birth length, infant mortality

	Linear Running Variable				Quadratic Running Variable			
	Main	CERRD	Baseline	Bias Corrected	Main	CERRD	Baseline	Bias Corrected
Panel A: Gestations weeks								
Birth weight <1500	-0.563	-0.761	-0.460	-0.563	-0.883*	-0.586	-0.773*	-0.883**
	(0.506)	(0.590)	(0.412)	(0.412)	(0.491)	(0.628)	(0.433)	(0.433)
Observations	416,913	416,913	416,913	416,913	416,913	416,913	416,913	416,913
Optimal Bandwidth	191.0	132.5	191.0	191.0	404.5	266.4	404.5	404.5
Panel B: Birth weight (grams)								
Birth weight <1500	-217.471**	-327.932***	-177.662**	-217.471***	-181.407**	-227.692*	-161.870*	-181.407**
	(101.057)	(125.384)	(84.129)	(84.129)	(92.070)	(122.305)	(84.529)	(84.529)
Observations	416,892	416,892	416,892	416,892	416,892	416,892	416,892	416,892
Optimal Bandwidth	162.8	113.0	162.8	162.8	361.2	237.8	361.2	361.2
Panel C: Birth Length (cms)								
Birth weight <1500	-0.340	-0.564	-0.217	-0.340	-0.422	-0.482	-0.329	-0.422
	(0.593)	(0.671)	(0.490)	(0.490)	(0.544)	(0.713)	(0.486)	(0.486)
Observations	416,886	416,886	416,886	416,886	416,886	416,886	416,886	416,886
Optimal Bandwidth	189.0	131.2	189.0	189.0	414.4	272.9	414.4	414.4
Panel D: Infant Mortality								
Birth weight <1500	0.006	-0.000	0.002	0.006***	0.009***	-0.001	0.008***	0.009***
	(0.004)	(0.004)	(0.001)	(0.001)	(0.002)	(0.003)	(0.002)	(0.002)
Observations	417,304	417,304	417,304	417,304	417,304	417,304	417,304	417,304
Optimal Bandwidth	140.5	97.5	140.5	140.5	273.8	180.3	273.8	273.8

Notes: Alternative specifications are documented for each intergenerational outcome considered in Table 2. Each panel considers a specific outcome, based on local linear (columns 1-4), and local quadratic models (columns 5-8). Alternative models consist of robust bias corrected models (column 1 and 5), models using Calonico et al. (2020)'s minimum error coverage bandwidth selection (columns 2 and 4), using standard RD models without robust bias correction (columns 3 and 5), and using bias corrected but not robust models (columns 4 and 8). Significance stars for two-sided test: * $p<0.10$; ** $p<0.05$; *** $p<0.01$.

Table D2: Alternative Specifications – Prematurity, VLBW, Fetal Growth, Sex ratio, Anderson Index

	Linear Running Variable				Quadratic Running Variable			
	Main	CERRD	Baseline	Bias Corrected	Main	CERRD	Baseline	Bias Corrected
Panel E: Prematurity								
Birth weight <1500	0.150**	0.198**	0.132**	0.150***	0.170**	0.155*	0.155***	0.170***
	(0.071)	(0.084)	(0.058)	(0.058)	(0.067)	(0.085)	(0.060)	(0.060)
Observations	417,304	417,304	417,304	417,304	417,304	417,304	417,304	417,304
Optimal Bandwidth	182.7	126.8	182.7	182.7	389.9	256.8	389.9	389.9
Panel F: Very low birth weight status								
Birth weight <1500	0.045***	0.029	0.042***	0.045***	0.047**	0.037	0.045***	0.047***
	(0.017)	(0.020)	(0.014)	(0.014)	(0.020)	(0.026)	(0.017)	(0.017)
Observations	417,304	417,304	417,304	417,304	417,304	417,304	417,304	417,304
Optimal Bandwidth	242.3	168.1	242.3	242.3	456.7	300.8	456.7	456.7
Panel G: Fetal growth rate								
Birth weight <1500	-4.426**	-6.831**	-3.721**	-4.426**	-3.438*	-5.080*	-3.094*	-3.438*
	(2.156)	(2.757)	(1.762)	(1.762)	(1.952)	(2.648)	(1.771)	(1.771)
Observations	416,892	416,892	416,892	416,892	416,892	416,892	416,892	416,892
Optimal Bandwidth	167.5	116.2	167.5	167.5	375.9	247.5	375.9	375.9
Panel I: Anderson Index								
Birth weight <1500	-0.415*	-0.423	-0.359*	-0.415**	-0.403*	-0.445	-0.375*	-0.403*
	(0.236)	(0.303)	(0.194)	(0.194)	(0.242)	(0.304)	(0.208)	(0.208)
Observations	417,304	417,304	417,304	417,304	417,304	417,304	417,304	417,304
Optimal Bandwidth	227.5	157.8	227.5	227.5	485.8	319.9	485.8	485.8

Notes: Refer to notes to Table D1. Identical models are presented, however considering alternative outcomes in each of panels E-I. Significance stars for two-sided test: * p<0.10; ** p<0.05; *** p<0.01.

Table D3: Identification Considerations – Donut RD and Placebo Treatments

	Gestation Length (1)	Birth weight (2)	Size (3)	Infant Mortality (4)	Premature (5)	VLBW (6)	Fetal Growth (7)
Panel A: Original Model							
Birth weight < 1,500	-0.563	-217.471**	-0.340	0.006	0.150**	0.045***	-4.426**
	(0.506)	(101.057)	(0.593)	(0.004)	(0.071)	(0.017)	(2.156)
Panel B: Donut RDD							
Donut = 0	-0.563	-217.471**	-0.340	0.006	0.150**	0.045***	-4.426**
	(0.506)	(101.057)	(0.593)	(0.004)	(0.071)	(0.017)	(2.156)
Donut = 5	-0.562	-225.410**	-0.388	0.005	0.139*	0.045***	-4.955**
	(0.549)	(108.272)	(0.643)	(0.004)	(0.073)	(0.017)	(2.283)
Donut = 10	-0.493	-225.994**	-0.336	0.005	0.112	0.045**	-5.297**
	(0.547)	(111.611)	(0.651)	(0.004)	(0.068)	(0.018)	(2.351)
Donut = 20	-0.098	-164.466**	-0.051	0.006	0.057	0.033**	-3.565**
	(0.333)	(76.287)	(0.457)	(0.005)	(0.053)	(0.013)	(1.733)
Panel C: Alternative Cut-offs							
Birth weight < 1,250	0.257	9.975	-0.491	-0.061***	-0.014	0.049	0.279
	(0.441)	(155.533)	(0.591)	(0.015)	(0.055)	(0.031)	(3.692)
Birth weight < 1,750	-0.077	-6.215	-0.139	-0.005	-0.000	0.005	-0.053
	(0.224)	(62.496)	(0.241)	(0.005)	(0.026)	(0.013)	(1.347)
Birth weight < 2,000	0.088	31.475	0.150	-0.003	-0.002	0.002	0.532
	(0.104)	(24.824)	(0.166)	(0.006)	(0.015)	(0.005)	(0.542)
Birth weight < 2,250	0.032	-1.857	0.029	-0.003	0.006	0.004	-0.169
	(0.061)	(22.627)	(0.116)	(0.005)	(0.010)	(0.004)	(0.529)
Birth weight < 2,500	-0.055	-40.489**	-0.097	0.002	0.011	0.004	-0.892**
	(0.058)	(17.015)	(0.084)	(0.002)	(0.010)	(0.004)	(0.358)

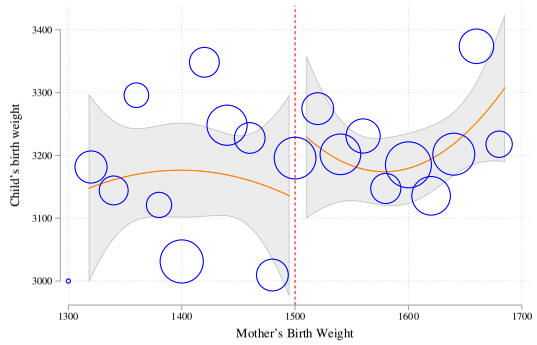
Notes: Panel A replicates estimates from Table 2 for ease of comparison. Panel B estimates RDD models following 1, however removing observations within x grams of the cut-off, where x is indicated in the Donut hole value. Panel C estimates RDD estimates following equation 1, however here rather than using the true discontinuity at 1,500 grams, alternative discontinuities at other maternal birth weight points are considered. All additional notes follow those to Table 2.

Table D4: Intensive Health Investments and Birth Outcomes of the Second Generation (All births)

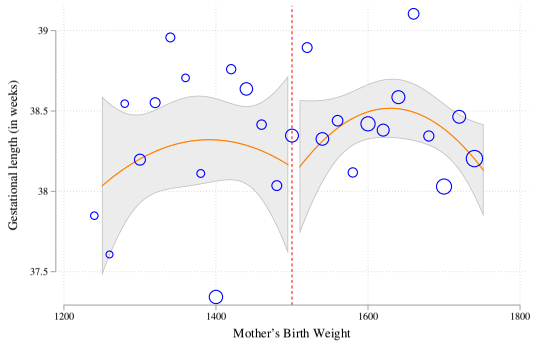
Panel A: Baseline Variables	Gestation (weeks) (1)	Birth weight (grams) (2)	Birth length (cms) (3)	Infant Mortality (4)
Birth weight < 1,500	-0.099	-109.043	-0.172	-0.011**
	(0.252)	(70.694)	(0.334)	(0.005)
	[0.653]	[0.939]	[0.696]	[0.020]
Mean of Dep. Var.	38.341	3192.751	48.682	0.008
Observations	418,855	418,834	418,827	419,248
Optimal Bandwidth	254.7	189.3	242.9	252.2
Effective Observations	1,878	1,304	1,741	1,880
Observations (left)	695	531	663	696
Observations (right)	1,183	773	1,078	1,184
q-sharpened p-value	0.248	0.110	0.248	0.098
Panel B: Transformed Measures	Prematurity (5)	Very low birth weight (6)	Fetal growth rate (7)	Anderson Index (8)
Birth weight < 1,500	0.089**	0.024**	-2.682	-0.114
	(0.043)	(0.009)	(1.660)	(0.153)
	[0.981]	[0.994]	[0.947]	[0.947]
Mean of Dep. Var.	0.093	0.013	82.740	-0.070
Observations	419,248	419,248	418,834	419,248
Optimal Bandwidth	166.7	204.4	193.8	258.3
Effective Observations	1,178	1,508	1,340	1,881
Observations (left)	477	602	547	697
Observations (right)	701	906	793	1,184
q-sharpened p-value	0.098	0.098	0.110	

Notes: Each column displays estimates of the change in the given dependent variable from above to below the 1,500 gram assignment threshold for mothers. In each case, local linear regression is used with a triangular kernel, calculating the MSE optimal bandwidth of Calonico et al. (2020). Robust bias corrected standard errors are reported in parentheses. Below standard errors, a one tailed t-test is calculated, which can be viewed as the support in favour of there actually being *positive* intergenerational transmission to the second generation. q-sharpened p-values refer to corrections conducted across the entire class of outcomes. * p<0.10; ** p<0.05; *** p<0.01.

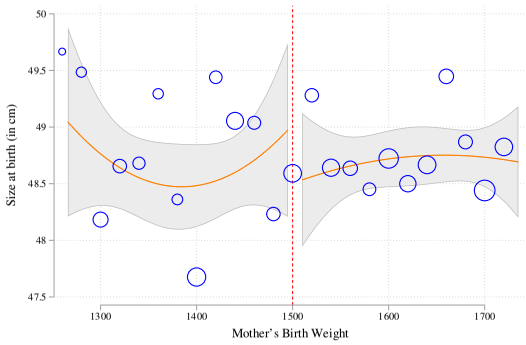
Figure D3: Descriptive Plots of Parental Policy Receipt and Child Health Measures (All births)



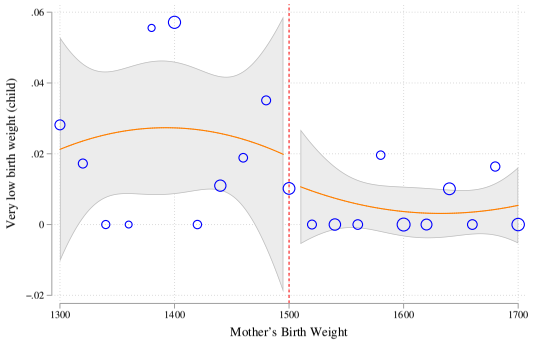
(a) Birth weight



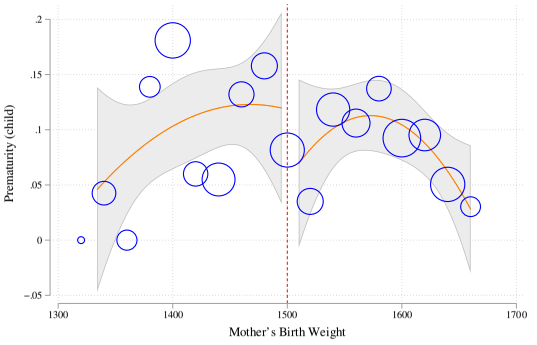
(b) Gestational period



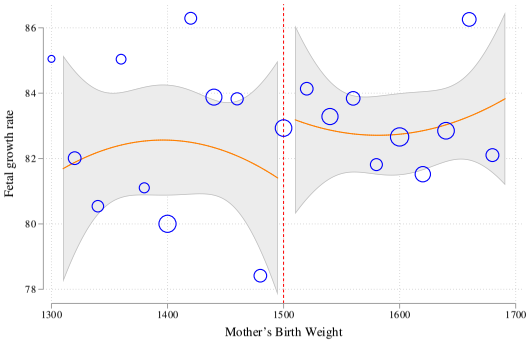
(c) Gestational length



(d) Very low birth weight



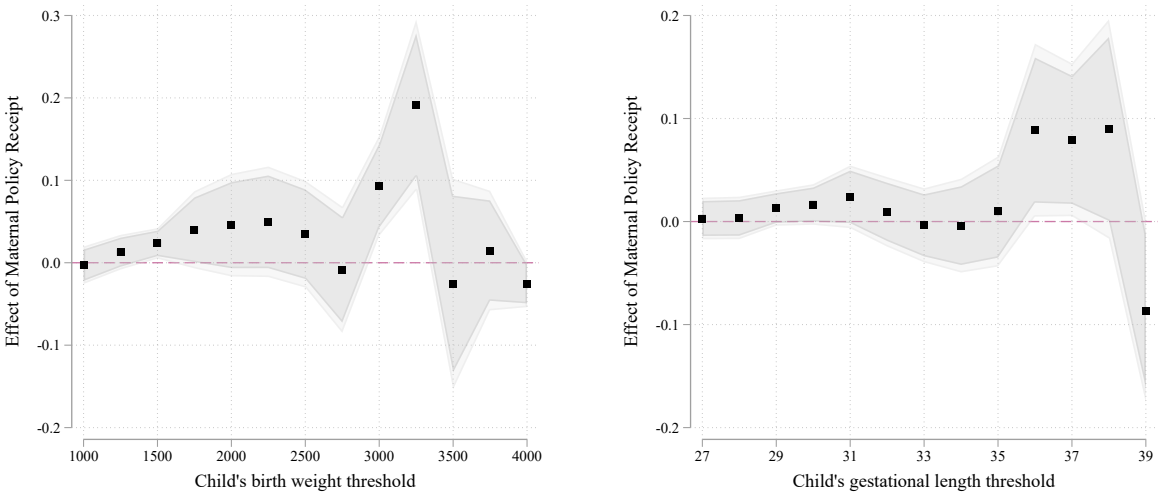
(e) Prematurity



(f) Fetal growth rate

Notes: Plots show separate quadratic fits estimated on each side of the 1,500 gram cut-off, in each case restricting attention to observations within the optimal bandwidth following Calonico et al. (2014). 95% confidence intervals of the quadratic fit are estimated, and circles represent average outcomes in 20 gram bins. Observations at 1,500 grams are plotted in binned averages, but are not used in estimating the quadratic fit.

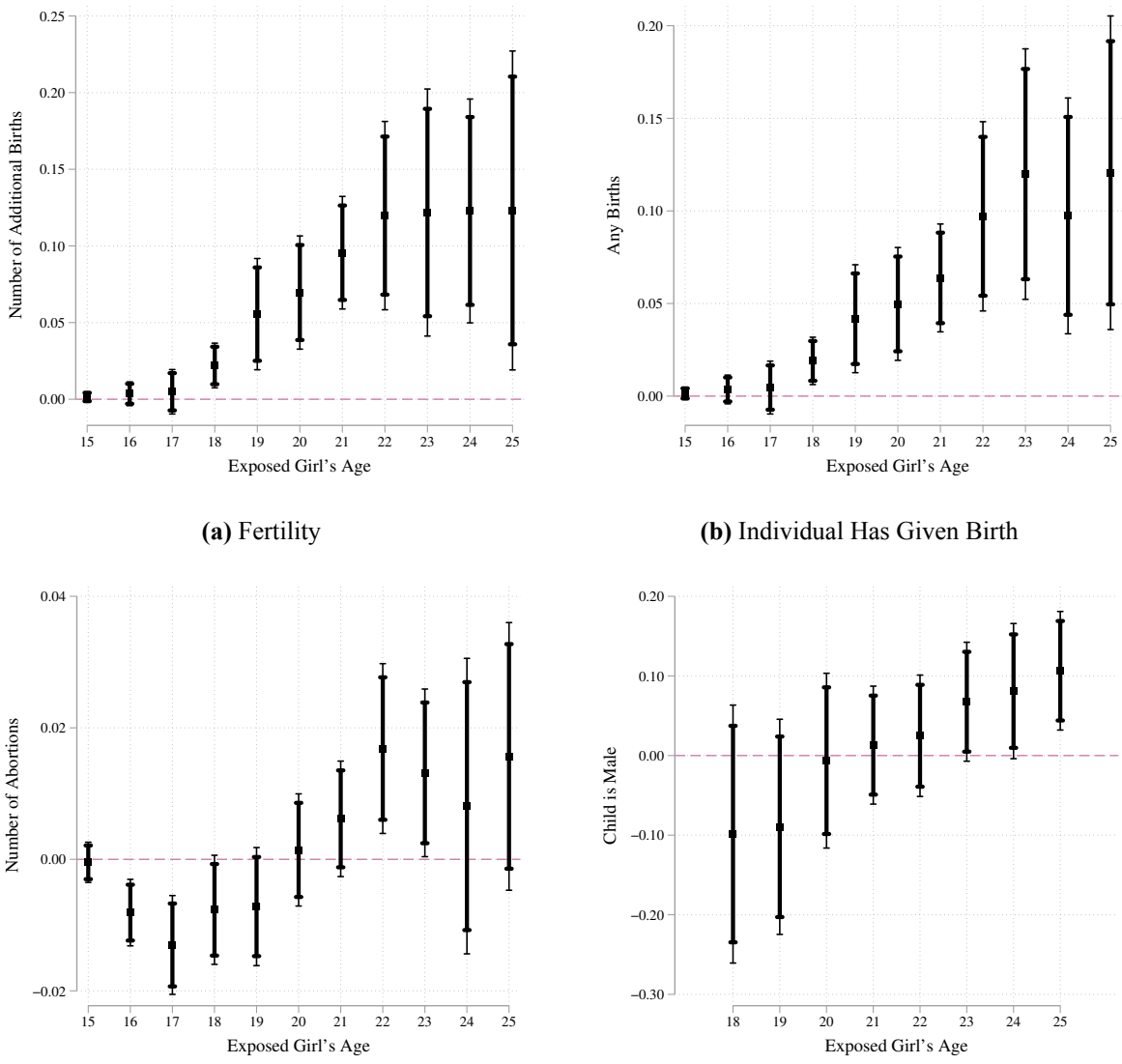
Figure D4: Distributional Impacts of Early Life Health Interventions on Second Generation Health Stocks (All births)



(a) Birth weight (All) **(b)** Gestational length (All)

Notes: Each point estimate (black square) and 90 and 95% confidence interval (dark and light shaded areas respectively) refers to RDD estimates of the likelihood that a birth occurring to a treated girl born has health stocks (birth or gestational weight) below the cut-off indicated on the x-axis of each sub-plot. Panels (a) and (c) refer to all individuals, while panel (b) and (d) refer to first generation individuals who were born at above 32 weeks of gestation, and are hence more clearly targeted by the policy.

Figure D5: Impacts of Early Life Health Interventions on Fertility and Spontaneous Abortions (All births)



(a) Fertility

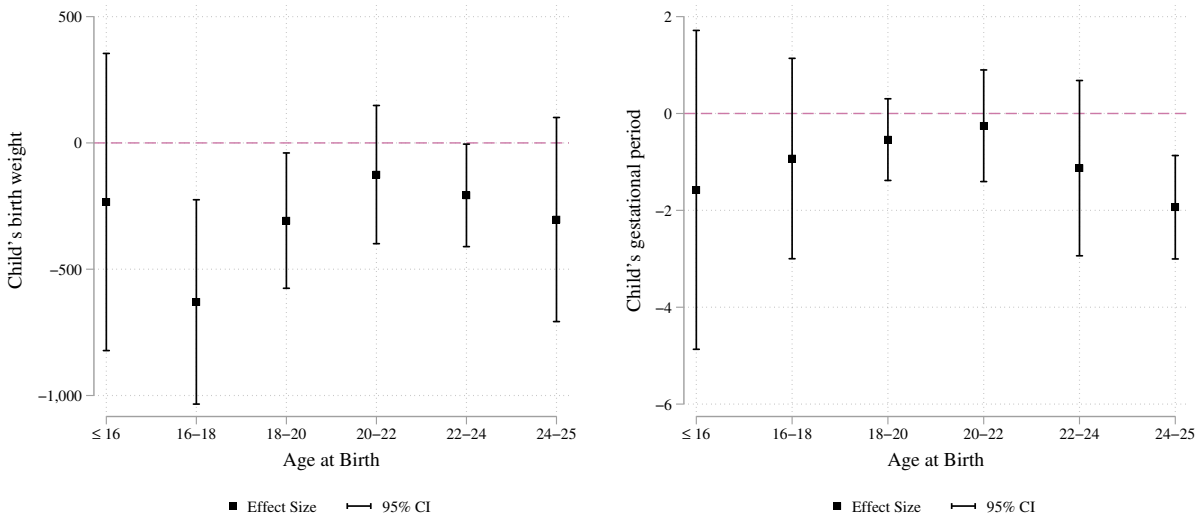
(b) Individual Has Given Birth

(c) Abortions

(d) Male Child

Notes: Each point estimate and confidence interval refer to the impacts of early life medical investment on the number of births an individual has had by each age (panel (a)), the probability an individual has had a birth by each age (panel(b)), the number of abortions observed in hospitalisation data (panel (c)), and the likelihood that an observed birth is a male (panel (d)). Thicker black error bars present 90% CIs, while thinner error bars report 95% CIs. All estimates follow the procedures laid out in section 4, and report RBC estimates using a local linear regression with a triangular kernel in the MSE optimal bandwidth.

Figure D6: Estimated RDD Effects by Second Generation Mother's Age



(a) Birth weight

(b) Gestational length

Notes: Each point estimate (black square) and 95% confidence interval refer to RDD estimates of the effect of treatment on birth weight (panel (a)) and gestational length (panel (b)), for second generation mothers are ages indicated in panels. Each estimate is generated using the same sample and methods described in Notes to Table 2, however sub-setting exclusively to the ages indicated on the horizontal axis.

Table D5: Intensive Health Investments and Birth Outcomes of the Second Generation in Adolescent Mothers (1992-2000 Cohorts)

Panel A: Baseline Variables	Gestation (weeks) (1)	Birth weight (grams) (2)	Birth length (cms) (3)	Infant Mortality (4)
Birth weight < 1,500	-0.441	-142.526	-0.196	0.007*
	(0.501)	(96.561)	(0.559)	(0.004)
	[0.810]	[0.930]	[0.637]	[0.957]
Mean of Dep. Var.	38.423	3158.790	48.586	0.007
Observations	416,268	416,247	416,242	416,657
Optimal Bandwidth	197.1	172.2	194.6	142.5
Effective Observations	881	805	879	678
Observations (left)	305	284	305	251
Observations (right)	576	521	574	427
q-sharpened p-value	0.325	0.248	0.406	0.206
Panel B: Transformed Measures	Prematurity (5)	Very low birth weight (6)	Fetal growth rate (7)	Anderson Index (8)
Birth weight < 1,500	0.133*	0.045***	-2.524	-0.335
	(0.072)	(0.016)	(1.996)	(0.223)
	[0.968]	[0.998]	[0.897]	[0.933]
Mean of Dep. Var.	0.086	0.015	81.851	-0.071
Observations	416,657	416,657	416,247	416,657
Optimal Bandwidth	190.6	259.3	181.0	248.8
Effective Observations	877	1,280	848	1,182
Observations (left)	305	361	294	352
Observations (right)	572	919	554	830
q-sharpened p-value	0.206	0.031	0.248	

Notes: Each column displays estimates of the change in the given dependent variable from above to below the 1,500 gram assignment threshold for mothers. In each case, local linear regression is used with a triangular kernel, calculating the MSE optimal bandwidth of Calonico et al. (2020). Robust bias corrected standard errors are reported in parentheses. Below standard errors, a one tailed t-test is calculated, which can be viewed as the support in favour of there actually being *positive* intergenerational transmission to the second generation. q-sharpened p-values refer to corrections conducted across the entire class of outcomes. * p<0.10; ** p<0.05; *** p<0.01.

Table D6: Effects Conditioning on Education and Partnership Choices

	Gestation Length (1)	Birth weight (2)	Size (3)	Infant Mortality (4)	Premature (5)	VLBW (6)	Fetal Growth (7)	Anderson Index (8)
Panel A: Controlling for Education								
Birth weight < 1,500	-0.520	-197.183**	-0.260	0.007*	0.143**	0.044***	-3.980**	-0.391*
	(0.486)	(93.490)	(0.554)	(0.004)	(0.069)	(0.016)	(1.983)	(0.220)
Mean of Dep. Var.	38.450	3159.384	48.619	0.007	0.085	0.015	81.919	-0.076
Observations	416,858	416,837	416,831	416,863	416,863	416,863	416,837	416,863
Optimal Bandwidth	194.2	165.0	195.4	151.6	183.2	251.0	169.6	233.8
Panel B: Conditioning on Highly Educated								
Birth weight < 1,500	-1.960**	-358.954**	-1.888**	0.012*	0.336***	0.092**	-5.292*	-0.703*
	(0.870)	(154.109)	(0.938)	(0.007)	(0.088)	(0.043)	(3.201)	(0.383)
Mean of Dep. Var.	38.488	3164.449	48.627	0.008	0.067	0.009	81.988	-0.032
Observations	225,863	225,855	225,851	225,865	225,865	225,865	225,855	225,865
Optimal Bandwidth	163.9	160.8	148.2	151.4	141.7	188.6	175.0	187.0
Panel C: Controlling for partner presence								
Birth weight < 1,500	-0.567	-213.094**	-0.365	0.006	0.149**	0.044***	-4.322**	-0.417*
	(0.501)	(102.405)	(0.595)	(0.004)	(0.070)	(0.017)	(2.178)	(0.241)
Mean of Dep. Var.	38.446	3160.307	48.609	0.006	0.085	0.013	81.919	-0.075
Observations	416,913	416,892	416,886	417,304	417,304	417,304	416,892	417,304
Optimal Bandwidth	186.9	164.2	182.7	133.2	183.5	240.0	169.0	220.0
Panel D: Conditioning on partner presence								
Birth weight < 1,500	-0.519	-106.399	-0.277	0.008*	0.112	0.035**	-2.139	-0.397
	(0.534)	(122.319)	(0.670)	(0.004)	(0.076)	(0.014)	(2.325)	(0.273)
Mean of Dep. Var.	38.402	3152.810	48.484	0.009	0.088	0.012	81.676	-0.084
Observations	353,216	353,200	353,192	353,475	353,475	353,475	353,200	353,475
Optimal Bandwidth	198.5	190.3	209.0	141.8	198.7	346.8	220.4	235.9

Notes: Results replicate estimates from Table 2 conditioning on mother's education at time of birth (panel A), or whether the father was observed to be present in the child's registration (panel C). Results condition on the mother coming from a highly educated (grandmother's education above the median) (panel B), or on the father being present in the child's registration (panel D). All additional notes follow those to Table 2.

E Additional Details: Selection into Treatment

E.1 Selection into Treatment in the First Generation

In the body of the paper, we focus on selection into second generation births which owe to selective fertility. However, an identical issue could arise if selection occurs due to survival at birth when intensive medical treatments were initially provided. As documented by Bharadwaj et al. (2013), and as discussed further in Appendix B.1, this policy resulted in selective survival of individuals who received more intensive treatment, and as such, likely saved individuals with poorer health stocks. In order to consider whether this initial selective survival can explain our observed results, we conduct a counterfactual experiment where we impute non-surviving individuals on the right hand side of the cut-off, and consider whether specific life courses for these individuals could explain away the negative observed intergenerational transmission.

The results of this exercise are provided in Table E1. This table makes clear that while selective survival of the first generation could affect the magnitude of the estimates, it would generally be insufficient to invert the sign of point estimates. For example, consider the case of child birth weight. If all counterfactual individuals on the right-hand side of the cut-off had a birth weight equal to that of the tenth percentile, the intergenerational effect could be as small as -163 grams. While if these non-surviving individuals had birth weight in the 90th percentile, the true effect could be as large as -281 grams. This exercise is particularly clear in the case of a child's low birth weight status. Regardless of how extreme we make the counterfactual scenarios, the true effect remains large, and statistically significant, even at 1%.

This analysis considers counterfactual stocks of children who did not exist in the second generation due to selective survival in the first generation. More generally, it is illustrative to consider the range of counterfactual scenarios over which it would be sufficient to turn negative observed intergenerational impacts into positive effects. In Table E1 we assume that all imputed individuals would have had fertility rates which were equal to those of surviving individuals in their birth weight bin.

Specifically, the exercise consists of the following. Using estimates of the policy impact on infant mortality, we calculate the number of individuals on the right-hand side of the birth weight cut-off who we estimate would have survived had they received the treatment. This is estimated precisely using the RDD models discussed in this paper. We then impute the proportion of these individuals who would have given birth by each age using 20 gram birth weight bin-specific actual averages observed of this variable. We then add these additional 'counterfactual' births to our second generation sample (note that given different optimal bandwidths, the number of imputed individuals will vary slightly by outcome). As we do not know what their health outcomes would have been at birth, we consider a range of scenarios, imputing health outcomes across the actual distribution of health at birth.

The results from this activity are plotted for birth weight (Figure E1a), gestational weeks (Figure E1b), size at birth (Figure E1c), and fetal growth rate (Figure E1d). Binary outcomes are not considered, as these often entirely omit extreme outcomes such as VLBW which occur in less than 5% of cases. Considering birth weight as the outcome of interest, it appears highly unlikely that selective survival at birth could explain the observed negative intergenerational transfers from mothers to children. For this to be the case, non-surviving individuals would have had to be highly fertile (each having 1 birth), while at the same time giving birth to children with very low health stocks (all at the 5th health percentile). Similar extreme patterns are observed for both gestational weeks and the fetal growth rate. In the case of size at birth, more feasible counterfactual outcomes exist which could explain away the negative intergenerational transmission (for example, average birth rates, and all counterfactual babies being born at the 20th health percentile or below. Nevertheless, across all outcomes considered, the evidence broadly suggests that correcting for survival at birth in the first generation would not be sufficient to turn around observed results in all health dimensions.

E.2 Compositional Effects and Fertility Selection

In the body of the paper we discuss a number of additional tests related to fertility and selection, and these are collected in this Appendix. These results are summarised below:

- Table E2: Reports the RD estimate where the outcome is the number of children born.
- Table E3: Corresponds to formal RDD estimates which examine whether the characteristics of parents of second generation children differ across the treatment threshold. These correspond to descriptive results presented in Figure 7 of the main paper.
- Figure E2: Reports marginal effects of health at birth by mother's age from descriptive regressions (left-hand panel) and mother fixed effect models (right-hand panels). Models consist of estimating:

$$Y_{ij} = \mu_{age} + \phi_j + \varepsilon_{ij}$$

  for child i born to mother j, where μ_{age} refers to mother age at birth fixed effects, and ϕ_j refers to mother fixed effects (included in right-hand column, but not left-hand column). Means along with their 99% confidence intervals are shown for each age fixed effect.

- Figure E3: Plots showing descriptive quantities for individuals marginally to the left (treated) and marginally to the right (untreated) of the cut-off. These show the mean number of children born by age (panel (a)), the probability of birth at a given age (panel (b)), whether an individual has had any births by a given age (panel (c)), and whether an individual has a birth at a given age (panel (d)). The difference between panels (a)/(c) and (b)/(d) is that the prior reports values *by* age (*i.e.* is cumulative), while the latter reports values *at* age (*i.e.* is not cumulative).
- Figure E4: Reports an identical plot to Figure E3, however around the median rather than around the treatment cut-off.
- Figure E5: Reports similar effects to those in Figure E2, but estimating separately by characteristics of exposed mothers or their families.
- Table E4: Balance tests where indicators for an individuals age at birth are regressed on treatment receipt within the bandwidth close to 1,500 grams ($\pm$ 245.5 grams, as per Table 2).

E.3 Additional Details: Marginal versus Average Effects in an RDD with Selection

In Section 5.3 we discuss what can be said about direct ('structural') effects of treatment receipt on health at birth of the following generation, and lay out a decomposition of estimate treatment effects in (7). To see how we arrive at (7), note that if we have two groups (the marginal mothers and the always mothers, we can write our estimated treatment effect as follows:

$$\begin{aligned} \alpha \;=\; & \left(E[Y_i^A|BW < 1500] - E[Y_i^A|BW \geq 1500]\right)(1-\mu) \\ & + \left(E[Y_i^M|BW < 1500] - E[Y_i^A|BW \geq 1500]\right)\mu \end{aligned} \tag{E1}$$

where μ refers to the portion of marginal mothers and $(1-\mu)$ the remaining portion of always-mothers. The key point to note here is that *because we do not observe* marginal mothers on the untreated side of the cut-off, we use the outcome among always mothers as the comparison group in both cases, and this is why $E[Y_i^A|BW \geq 1500]$ appears two times.

Now, if we add and subtract $\mu E[Y_i^M|BW < 1500]$, and rearrange, we can write the estimated treatment effect as follows:

$$\begin{aligned} \alpha &= \left(E[Y_i^A|BW < 1500] - E[Y_i^A|BW \geq 1500]\right)(1-\mu) \\ &\quad + \left(E[Y_i^M|BW < 1500] - E[Y_i^M|BW \geq 1500]\right)\mu \\ &\quad + \left(E[Y_i^M|BW \geq 1500] - E[Y_i^A|BW \geq 1500]\right)\mu. \end{aligned} \tag{E2}$$

In the paper we refer to the quantity $E[Y_i^M|BW \geq 1500] - E[Y_i^A|BW \geq 1500]$ as Δ (the difference between births occurring to marginal and always treated mothers in the absence of treatment). Thus, we can re-write (E3) as:

$$\alpha = \alpha^A(1-\mu) + \alpha^M\mu + \Delta\mu, \tag{E3}$$

which is the decomposition provided in the paper.

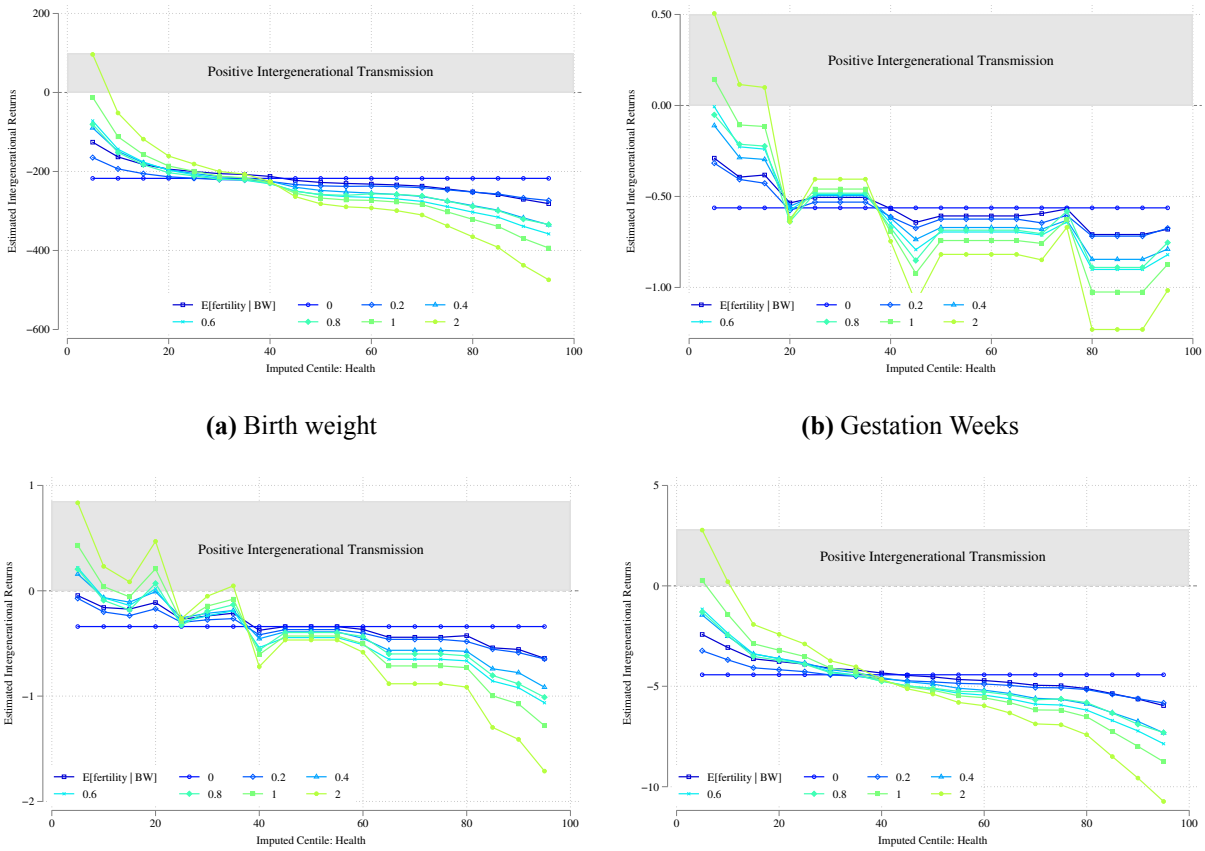
We present a graphical presentation of this decomposition in Figure E6. This also makes clear why an RDD estimate with selection may be negative even if the direct policy effect is positive in both groups. Namely, as laid out in Figure E6, if we do not observe outcomes for the selected group on one side of the cut-off (as indicated by the dashed blue line), and if these are instead compared to outcomes for a group which is positively selected, our estimated effect may be substantially different to a weighted average of the direct structural effects in each group.

Table E1: Selective survival and second generation outcomes – counterfactual analysis

		Counterfactual percentile assumed for survival				
	Baseline	10th	30th	50th	70th	90th
Panel A: Child's gestational length						
Birth weight < 1,500	-0.5626	-0.3936	-0.5051	-0.6070	-0.5939	-0.6801
	(0.5064)	(0.5005)	(0.5053)	(0.5098)	(0.5076)	(0.5162)
Effective Obs.	863	900	900	900	931	929
Imputed Obs.	0	72	72	72	75	75
Panel B: Child's birth weight						
Birth weight < 1,500	-217.47**	-163.13	-205.49**	-227.78**	-237.11**	-281.39***
	(101.06)	(99.50)	(101.11)	(101.86)	(102.14)	(103.70)
Effective Obs.	763	819	819	819	819	819
Imputed Obs.	0	64	64	64	64	64
Panel C: Child's birth size						
Birth weight < 1,500	-0.3395	-0.1590	-0.2388	-0.3402	-0.4412	-0.6414
	(0.5932)	(0.5868)	(0.5902)	(0.5939)	(0.5978)	(0.6049)
Effective Obs.	836	900	900	900	900	931
Imputed Obs.	0	72	72	72	72	75
Panel D: Child's infant mortality						
Birth weight < 1,500	0.006	0.006	0.006	0.006	0.006	0.006
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Effective Obs.	669	715	715	715	715	715
Imputed Obs.	0	50	50	50	50	50
Panel E: Child premature						
Birth weight < 1,500	0.1495**	0.1557**	0.1557**	0.1557**	0.1557**	0.0263
	(0.0706)	(0.0711)	(0.0711)	(0.0711)	(0.0711)	(0.0638)
Effective Obs.	834	898	898	898	898	931
Imputed Obs.	0	72	72	72	72	75
Panel F: Child VLBW						
Birth weight < 1,500	0.0449***	0.0439***	0.0439***	0.0439***	0.0439***	0.0439***
	(0.0169)	(0.0169)	(0.0169)	(0.0169)	(0.0169)	(0.0169)
Effective Obs.	1,164	1,255	1,255	1,255	1,255	1,255
Imputed Obs.	0	101	101	101	101	101
Panel G: Fetal growth rate						
Birth weight < 1,500	-4.4259**	-3.0703	-4.1083*	-4.5313**	-4.9493**	-5.9604***
	(2.1562)	(2.1313)	(2.1580)	(2.1652)	(2.1709)	(2.1948)
Effective Obs.	766	822	821	822	822	822
Imputed Obs.	0	64	64	64	64	64

Notes: Each panel displays outcomes under different counterfactual assumptions related to future outcomes for individuals who selectively did not survive birth as response to not receiving intensive treatment in generation 1. The left-hand panel replicates original estimates of policy receipt on intergenerational outcomes, and then additional columns impute outcomes for observations who selectively did not survive birth on the right-hand side of the treatment cut-off, assuming counterfactual outcomes at different percentiles of the health distribution at birth.

Figure E1: Intergenerational Transmission Under Alternative Health and Fertility Counterfactuals



(a) Birth weight

(b) Gestation Weeks

(c) Size at birth

(d) Fetal growth rate

Notes: Each point represents estimates of the impact of intensive health receipt at first generation births on the second generation health outcomes indicated as plot labels. Each estimate comes from a separate RD model based on all surviving observations, as well as imputed outcomes for individuals estimated to have not survived on the right-hand side of the birth weight cut-off due to lack of policy receipt. Imputations are made varying health outcomes (indicated on the horizontal axis), and fertility (indicated in the legend). All estimation details of each RD model follow those described in Table 2.

Table E2: Impacts of Early Life Health Interventions on Number of Children

Birth weight < 1,500	0.048***
	(0.015)
Observations	1,445,711
Mean of Dep. Var.	0.214
Optimal Bandwidth	177.0
Effective Observations	3,671
Mean of Dep. Var. (left)	0.235
Observations (left)	1,185
Mean of Dep. Var. (right)	0.204
Observations (right)	2,486

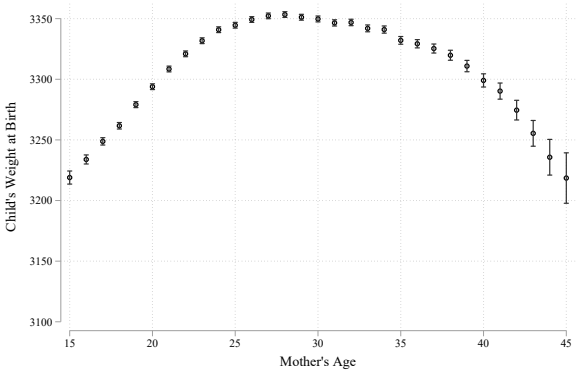
Notes: The table displays estimates of the change in the number of births from above to below the 1,500 gram assignment threshold. Local linear regression is used with a triangular kernel, calculating the MSE optimal bandwidth of Calonico et al. (2020). Robust bias corrected standard errors are reported in parentheses. * p<0.10; ** p<0.05; *** p<0.01.

Table E3: Is there Policy-Driven Selection into Childbirth? RD Estimates

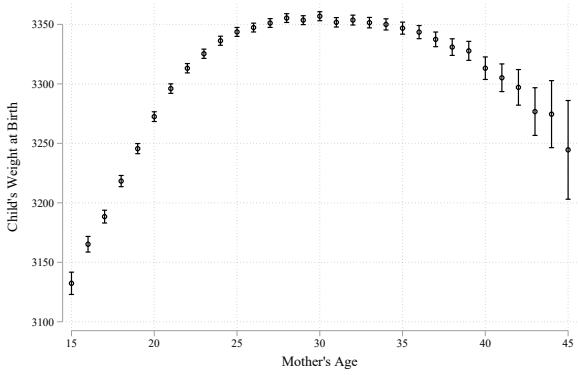
	Mother			Father				Parent's	Observed	Urban
	Age	Education	Employed	Age	Education	Employed	Married	Age Diff.	Dad	Status
Birth weight < 1,500	-0.447	-0.708*	-0.052	-3.757***	-0.622	0.114*	0.075*	-2.859***	-0.028	-0.087
	(0.318)	(0.409)	(0.055)	(0.906)	(0.386)	(0.063)	(0.040)	(1.092)	(0.070)	(0.060)
	[0.080]	[0.958]	[0.827]	[0.000]	[0.946]	[0.035]	[0.031]	[0.004]	[0.656]	[0.925]
Mean of Dep. Var.	19.910	11.228	0.185	23.373	11.492	0.636	0.072	3.644	0.842	0.874
Observations	431,164	430,708	431,166	363,912	363,639	431,166	431,166	363,912	431,166	431,166
Optimal Bandwidth	208.6	168.2	282.4	159.2	194.7	193.5	277.7	171.0	181.3	180.1
Effective Observations	1,016	786	1,490	635	752	893	1,411	690	859	859
Observations (left)	326	276	383	223	262	306	372	245	295	295
Observations (right)	690	510	1,107	412	490	587	1,039	445	564	564
q-sharpened p-value	0.190	0.155	0.218	0.001	0.168	0.155	0.155	0.042	0.249	0.190

Notes: Each column displays estimates of the change in the given dependent variable from above to below the 1,500 gram assignment threshold for newborns. In each case, local linear regression is used with a triangular kernel, calculating the MSE optimal bandwidth of Calonico et al. (2020). Robust bias corrected standard errors clustered at the gram level are reported in parentheses. p-values for one-sided tests are shown in square brackets, where in each case the null hypothesis is that there is a negative policy impact. Significance stars for two-sided test: * p<0.10; ** p<0.05; *** p<0.01.

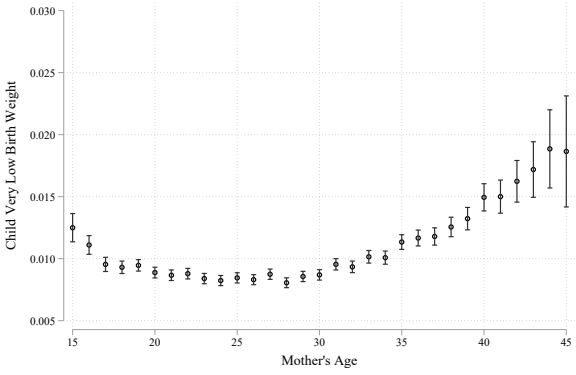
Figure E2: Weight at Birth by Mothers' Age



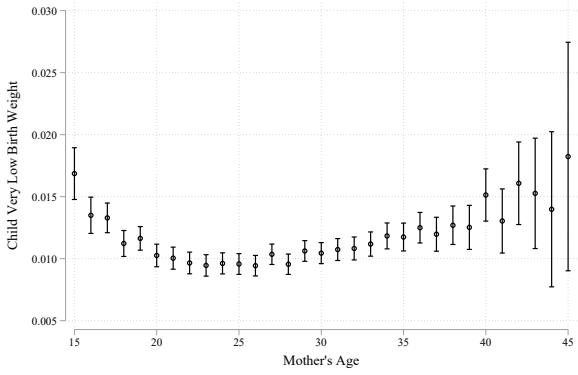
(a) Maternal age and child's birth weight



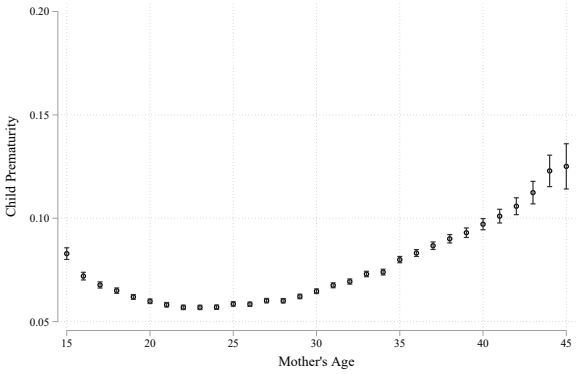
(b) Maternal age and child's birth weight – Mother FE



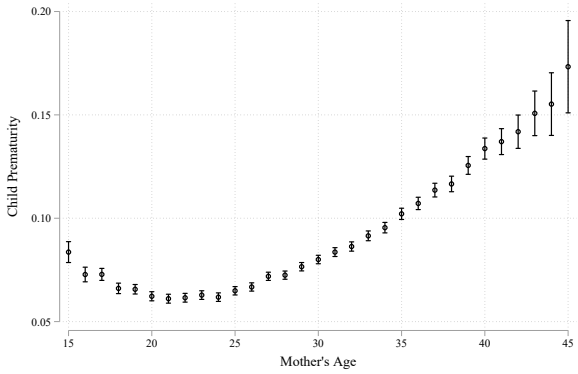
(c) Maternal age and child's VLBW status



(d) Maternal age and child's VLBW status – Mother FE



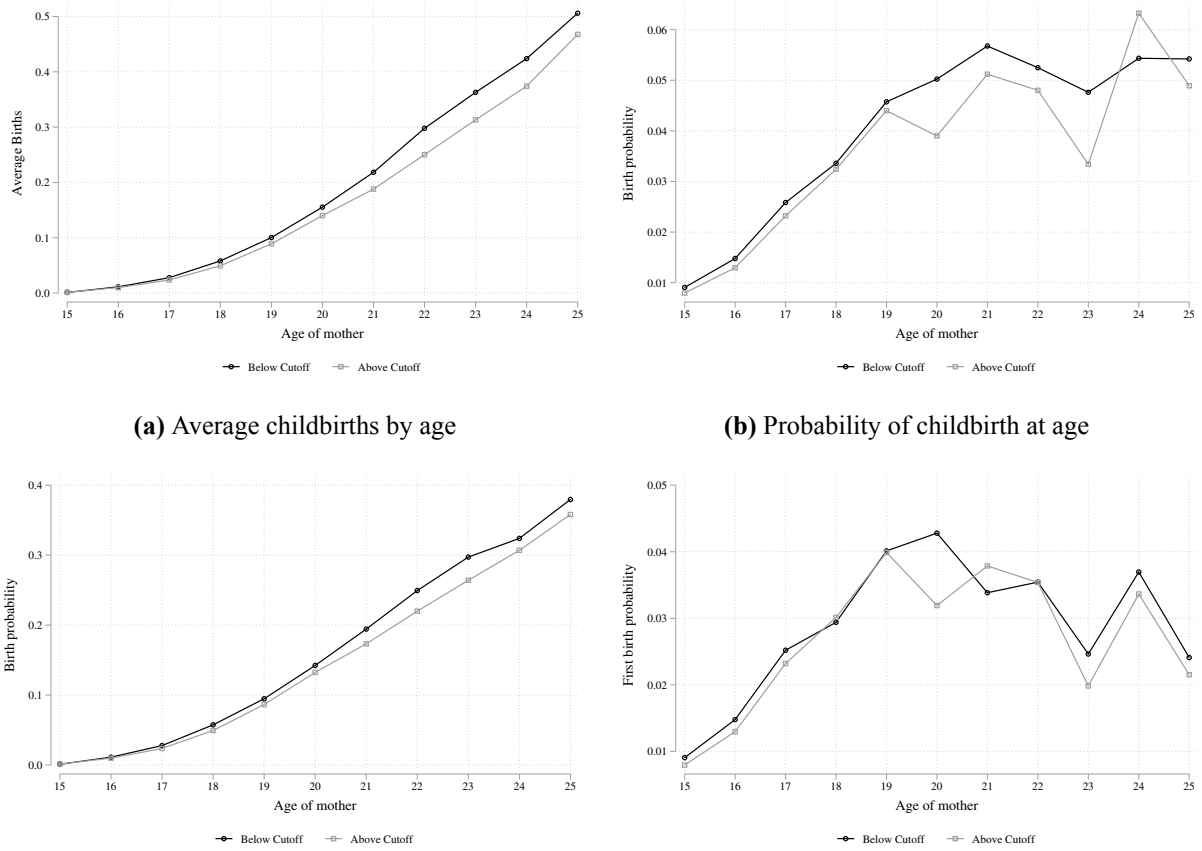
(e) Maternal age and child's prematurity



(f) Maternal age and child's prematurity – Mother FE

Notes: Each sub-plot documents average outcomes of individuals born to mothers whose age is indicated on the horizontal axis. Models are estimated with all observations in our period under study and include age fixed effects descriptively (left panel), or age fixed effects in a mother fixed effect model (right panel). Panels (a) and (b) consider the birth weight of children, panels (c) and (d) consider the average proportion of children who are very low birth weight (weight $<$ 1,500 grams), while panels (e) and (f) consider the average proportion of children who are premature (gestation $<$ 37 weeks). In the case of mother fixed effect models, marginal effects are documented for each age fixed effect.

Figure E3: Childbirth Around the 1,500 Grams Threshold



(a) Average childbirths by age

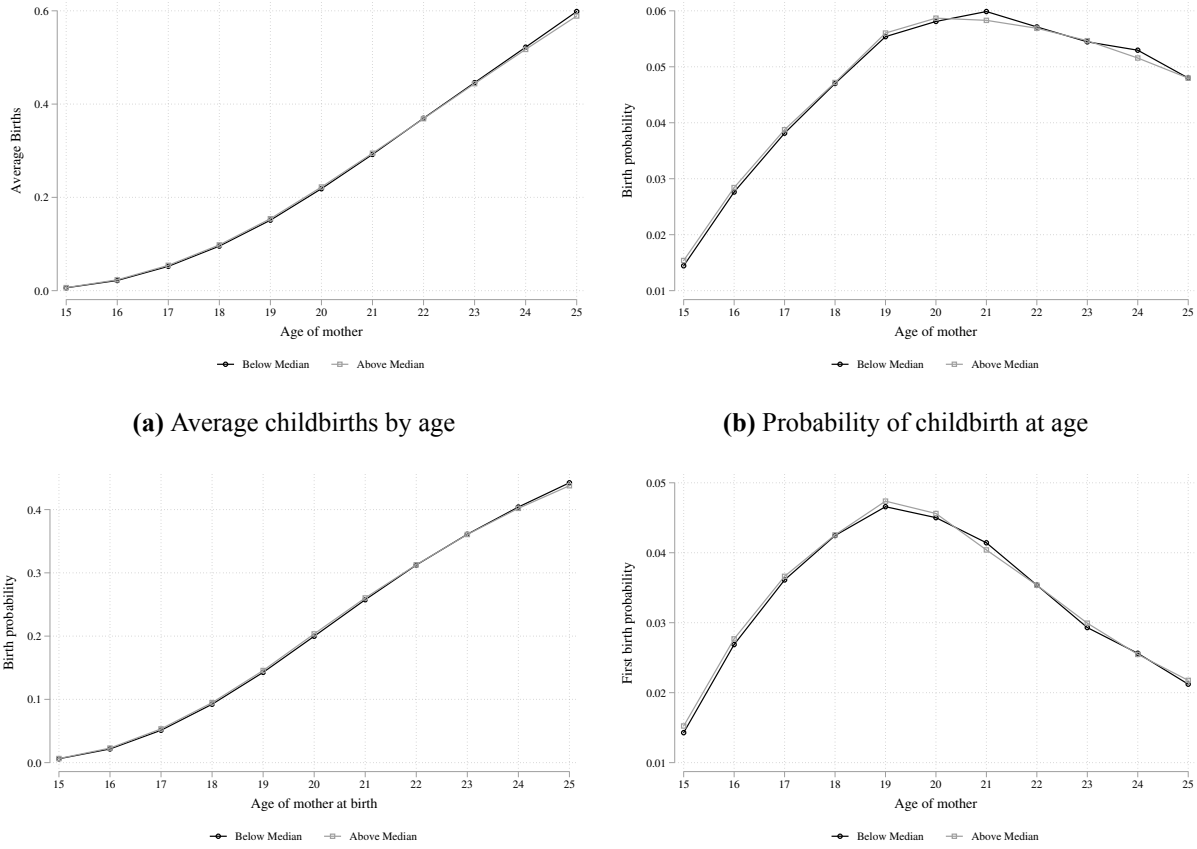
(b) Probability of childbirth at age

(c) Probability of childbirth by age

(d) Probability of first childbirth at age

Notes: Each point shows the estimated average number of childbirths by the age indicated (panel (a)), the probability that a mother has a birth at the age indicated (panel (b)) and by the age indicated (panel (c)), and the probability that a mother has the first birth at the age indicated (panel (d)). This is relative to the 1,500 grams cut-off and based on the 134.4 gram optimal bandwidth cut-off when considering infant mortality for the first generation.

Figure E4: Probability of Childbirth Relative to Median Birth Weight



(a) Average childbirths by age

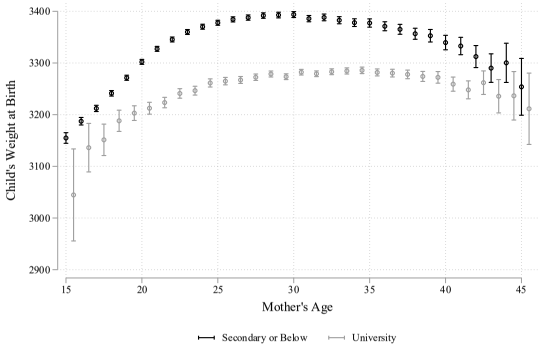
(b) Probability of childbirth at age

(c) Probability of childbirth by age

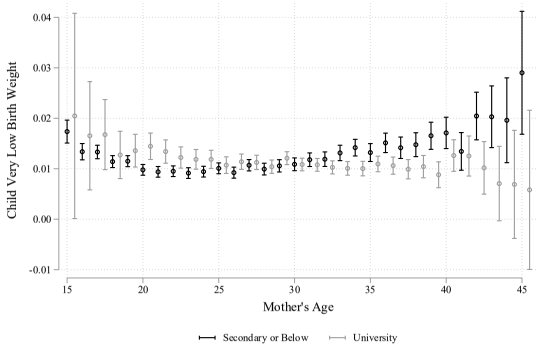
(d) Probability of first childbirth at age

Notes: Each point shows the estimated average number of childbirths by the age indicated (panel (a)), the probability that a mother has a birth at the age indicated (panel (b)) and by the age indicated (panel (c)), and the probability that a mother has the first birth at the age indicated (panel (d)). This is relative to the median of 3,300 grams.

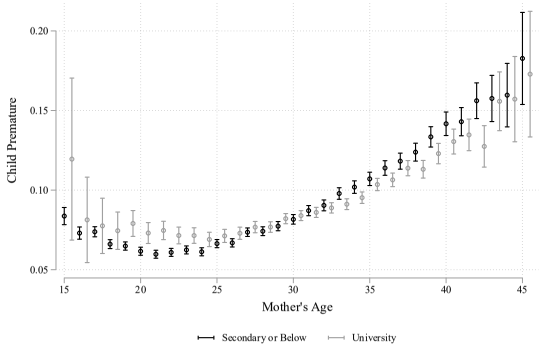
Figure E5: Weight at Birth by Mother's Age and Parent Characteristics



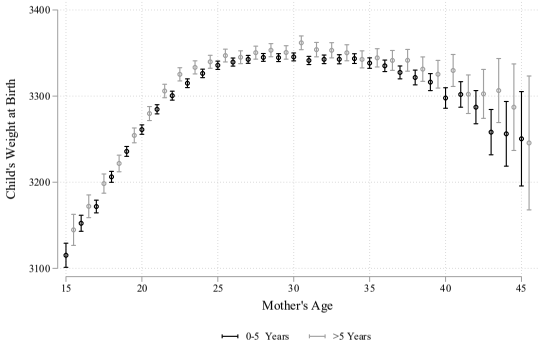
(a) Maternal age and children's birth weight



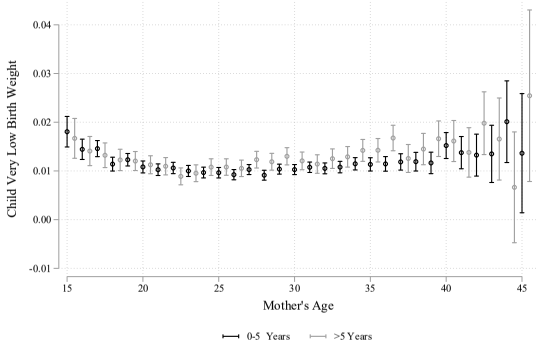
(b) Maternal age and children's very low birth weight



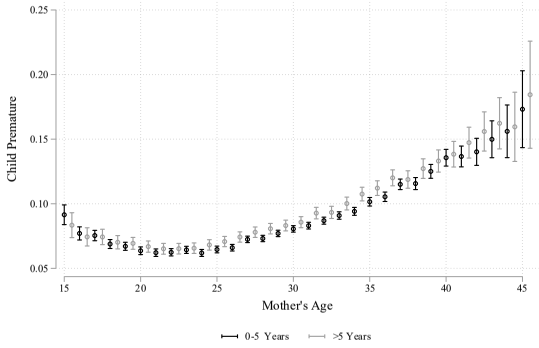
(c) Maternal age and children's premature



(d) Maternal age and children's birth weight



(e) Maternal age and children's very low birth weight



(f) Maternal age and children's premature

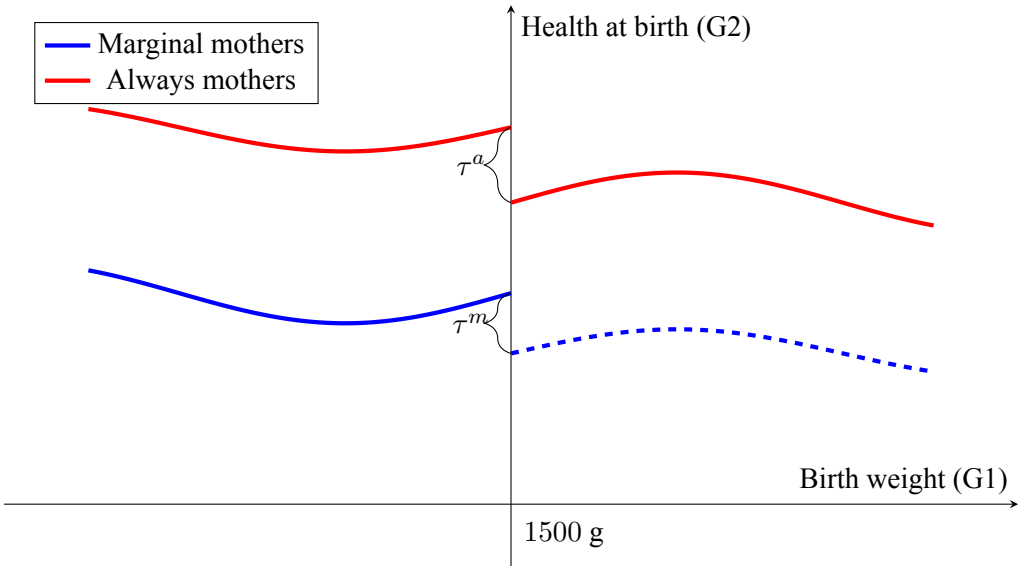
Notes: Each sub-plot documents by group the outcomes of individuals born to mothers whose age is indicated on the horizontal axis including mother-level fixed effects. Panels (a) and (d) consider the birth weight of second-generation children, panels (b) and (e) consider the average proportion of second-generation children who are very low birth weight (weight $<$ 1,500 grams), while panels (c) and (f) consider the average proportion of second-generation children who are premature (gestation $<$ 37 weeks). Optimally spaced bins and their 99% confidence intervals are documented as black points and error bands.

Table E4: Mother's Age Baseline Test by Family Characteristics – Generation 2

		Mother's Education		Grandmother's Education		Parents' Age Difference		Observed Father	
	All	Below median (12 years)	At or above median	Below median (9 years)	At or above median	Up to 5 years	More than 5 years	Observed	Non Observed
Mother's age: 15	-0.007	-0.029	0.000	-0.041**	0.026*	-0.010	-0.001	-0.003	-0.008
	(0.012)	(0.028)	(.)	(0.019)	(0.015)	(0.014)	(0.022)	(0.033)	(0.013)
Mother's age: 16	-0.009	-0.042	0.000	-0.022	0.003	-0.028	0.031	0.048	-0.019
	(0.015)	(0.035)	(.)	(0.022)	(0.022)	(0.018)	(0.029)	(0.045)	(0.016)
Mother's age: 17	0.006	-0.006	0.001	0.001	0.011	0.015	-0.012	-0.061	0.019
	(0.019)	(0.038)	(0.018)	(0.027)	(0.028)	(0.023)	(0.036)	(0.054)	(0.021)
Mother's age: 18	-0.001	0.000	-0.012	-0.026	0.024	-0.014	0.029	0.070	-0.013
	(0.021)	(0.038)	(0.023)	(0.031)	(0.029)	(0.026)	(0.038)	(0.057)	(0.023)
Mother's age: 19	0.030	0.011	0.048	0.009	0.052	0.018	0.056	-0.025	0.041*
	(0.023)	(0.033)	(0.031)	(0.031)	(0.033)	(0.027)	(0.040)	(0.062)	(0.024)
Mother's age: 20	0.005	0.003	0.015	0.026	-0.017	0.005	0.006	0.020	0.002
	(0.021)	(0.027)	(0.031)	(0.032)	(0.029)	(0.026)	(0.038)	(0.050)	(0.024)
Mother's age: 21	-0.014	0.025	-0.035	-0.010	-0.020	0.003	-0.054	-0.021	-0.014
	(0.021)	(0.025)	(0.031)	(0.031)	(0.029)	(0.026)	(0.036)	(0.052)	(0.023)
Mother's age: 22	-0.017	0.008	-0.025	0.012	-0.046	-0.019	-0.014	0.018	-0.023
	(0.019)	(0.018)	(0.030)	(0.026)	(0.028)	(0.023)	(0.035)	(0.054)	(0.021)
Mother's age: 23	-0.009	-0.002	-0.009	0.020	-0.036	-0.003	-0.020	-0.017	-0.008
	(0.016)	(0.019)	(0.024)	(0.020)	(0.024)	(0.019)	(0.029)	(0.036)	(0.018)
Mother's age: 24	0.002	0.023*	-0.005	0.013	-0.008	0.008	-0.012	-0.008	0.002
	(0.016)	(0.012)	(0.025)	(0.021)	(0.023)	(0.020)	(0.026)	(0.026)	(0.018)
Mother's age: 25	0.014	0.010	0.023	0.018	0.010	0.024	-0.009	-0.021	0.020
	(0.013)	(0.011)	(0.020)	(0.017)	(0.019)	(0.017)	(0.018)	(0.020)	(0.015)
Observations	1,175	479	696	574	601	783	392	194	981

Notes: Each column displays estimates of the change in the probability of a mother gives birth at a certain age from above to below the 1,500 gram assignment threshold in the given sub-sample. Significance stars for two-sided test: * $p<0.10$; ** $p<0.05$; *** $p<0.01$.

Figure E6: Understanding Regression Discontinuity Estimates With Sample Selection



F A Back-of-the-Envelope Estimation of Policy Costs and Benefits

Results from sections 5.1-5.2 suggest that the marginal impacts of this intensive early life medical care may actually act to *increase* costs levied in the second generation. This is at odds with the short-term marginal returns documented in previous literature pointing to greater survival (Almond et al., 2010), increased educational performance (Bharadwaj et al., 2013), and reduced links to state support programmes (Chyn et al., 2021). It is illustrative to consider what these second generation results imply vis-à-vis the policy's cost, and positive short term returns. The importance of considering such life-cycle returns to social policies has been espoused in García et al. (2020), and here we can move beyond the life cycle of the first generation, into second generation outcomes.

To do this, we consider estimates of the initial costs of providing marginal medical care to treated individuals, the estimated present value of first generation benefits of policy receipt, and the estimated present value of second generation costs. We incorporate into these calculations the welfare cost of financing taxation, and consider a money metric to compare across different policy domains. In each case, we are fortunate to have a directly matched control group to the treated group of individuals impacted by the policy. Namely, the counterfactual outcome for children treated at birth are children who narrowly miss out given the birth weight assignment rule. Nevertheless, we note that this is simply a back-of-the envelope activity, principally to allow us to determine how important the negative intergenerational transmission observed here may be compared with first generation benefits. Below we lay out estimated present values for costs and benefits along with any necessary assumptions to arrive to these figures (calculations are summarised in Appendix Table F1). This exercise is conducted from the point of view of a single marginally treated first generation child who is born in a public hospital, starting from the age of 0, with all costs and benefits discounted to the present day (and expressed in terms of real costs in 2022). In closing this sub-section, we note a number of elements which are likely affected by the initial medical care receipt, but which are not feasible to reduce to a money metric, and which are hence omitted from the calculation of policy returns.

The most obvious programme cost which must be covered by public funds are the costs of initial medical care. In the US, Almond et al. (2010) estimate that the cost of marginal medical care provided at this cut-off is 9,450 USD. In order to consider the costs in the context of Chile, we calculate the full costs associated with marginal estimated changes in hospitalisation. The additional use of hospitalisation days is documented in Figure 4a, at approximately 3.5 days in the first year of life, 1.2 days in year 2, 0.7 in year 3 and 0.5 in year 4 (after which no marginal changes in hospitalisation days are observed). Figures estimated from Chile related to the *total* cost of intensive care days (including medical inputs and care) suggest a value of 480,047 CLP in 2011 (Alvear et al., 2013), or approximately 694,000 CLP in current terms, which is equivalent to 855 USD per day.[23] Taking the net present value of costs to the health system at the time of birth, discounted at a 5% discount rate, this gives expected costs of 4,649 USD over the first years of the child's life. If we additionally incorporate the welfare cost of taxation noting the deadweight loss associated with the collection of tax revenues (Feldstein, 1999), suggested by García et al. (2020) as 50 cents on the dollar, this suggests an initial policy cost of 6,974 USD.

Early medical life receipt has been documented to be associated with a number of benefits. We consider here a reduced metric of first generation labour market outcomes, mapping from educational benefits documented by Bharadwaj et al. (2013). Based on their estimated impacts of policy receipt on education in the first generation, and a back of the envelope calculation of the labour market returns to education, they

[23] Note that Almond et al. (2010) estimate a cost of marginal treatment at birth of 9,450 USD in hospital costs in 2011 USD. Here, based on our estimate of 3.6 additional days of hospitalisation at birth, the equivalent instantaneous cost in USD of hospitalisation in Chile is $855 \times 3.6 = 3078$. Even when incorporating the welfare cost of taxation suggested by García et al. (2020), this value corresponds to 4,617 in 2022 USD, or 3,661 in 2011 USD. Thus costs in Chile are around a third of those in the US, in line with relative cost and hospitalisation usage indexes in the two countries (Lorenzoni and Koechlin, 2017).

suggest that marginal receipt of intensive health investments at birth may increase incomes by 2.7%. If we discount the expected flow of future earnings back to birth, based on the median income in Chile in 2021, this suggests directly attributable changes in salaries of 980 USD. Thus, labour market concerns alone do not cover the initial outlay, but if one additionally accounts for the value of a statistical life (VSL), and the marginal likelihood that individuals die during their first year of life, this suggests substantial additional benefits of 91,760 USD (Mardones and Riquelme, 2018).

In addition, we consider the associated costs to the reduction in health at birth of the second generation. We observed three main cost sources: an increase in the probability of receiving treatment in the second generation, an increase in infant mortality, and a decrease in birth weight. We calculate the net present value of the costs and benefits at the time of the mother's birth considering 0.24 additional births up to 25 years documented in Figure 6 for those mothers who were initially treated and the welfare cost of financing taxation. Taking the increase of 0.045 in the probability of being born at less than 1,500 grams and therefore of receiving the treatment, the net present value of the initial medical care costs is 22 USD. Due to the increase of 0.006 observed in the infant mortality rate for those born to treated mothers, a cost of 1,573 USD associated with the value of statistical lives lost must also be included. Finally, we consider changes in future income through changes in schooling. Educational benefits documented by Behrman and Rosenzweig (2004) indicate that increasing the birth weight by 1 lb. increases adult earnings by more than 7%. Thus, the 217 gram decrease in birth weight is associated with an additional cost of 86 USD. The above suggests a second generation cost of the policy of 1,682 USD. In contrast, we observe a decrease in hospitalisations during the first years of life which corresponds to an expected second generation benefit of 60 USD.

These calculations suggest that the total estimated cost of the policy amounts to 8,656 USD while the estimated benefit of the policy is 92,800 USD. If we only consider the 980 USD corresponding to labour returns to education received by the first generation and the 60 USD due to the decrease in early hospitalisation for the second generation, the amount is not enough to compensate the initial costs of medical care and the reduction in health of the second generation. However, when incorporating the marginal willingness to pay for the reduction of infant mortality through the VSL, the benefits are around 15 times the total costs. It is important to note that while we can provide a back-of-the-envelope estimate of the policy's net present value, this is necessarily based on a partial picture of the policy's full set of benefits. While we can feasibly value the policy's impact on health, education and future labour market outcomes, there are a number of benefits which we cannot easily quantify. This includes factors such as reduced rates of bereavement given lower rates of infant mortality and potential lower rates of pregnancy loss in the second generation,[24] as well as the policy's impact on allowing women and families greater autonomy to achieve their desired fertility.

It is important to note that due to a lack of recorded father's identifier, our estimated second generation costs and benefits are all based on mother–child links. In general, there is evidence to suggest significant intergenerational links between fathers and their children (Giuntella et al., 2019, for example). The calculations here all refer to costs and benefits for a single individual, and so are not affected if impacts are identical for fathers and mothers. The literature suggests that links between fathers and their children are generally smaller than those between mothers and their children (Chang et al., 2024), which, if anything, suggests that second generation costs and benefits would be lower for fathers. Given that costs exceed benefits in the second generation, all told it seems that the inclusion of fathers would only increase the total present value of policy receipt.

[24] Both of these events have obvious and considerable costs to well-being over myriad dimensions (Rogers et al., 2008; Ogwulu et al., 2015; Persson and Rossin-Slater, 2018). However adequately capturing their true value would be difficult.

Table F1: Present value of costs and benefits of policy

	USD	Source
Panel A: First Generation		
Costs		
Medical care per day: 855 USD		Alvear et al. (2013)
Year 1: 3.5 days	2,992.5	
Year 2: 1.2 days	1,026.0	
Year 3: 0.7 days	598.5	
Year 4: 0.5 days	427.5	
PV at birth	4,649.3	
Welfare cost of taxation	2,324.7	García et al. (2020)
Total	**6974.0**	
Benefits		
Labour market returns to education	979.7	Bharadwaj et al. (2013)
Value of a statistical life	91,760	Mardones and Riquelme (2018)
Total	**92,739.7**	
Panel B: Second Generation		
Costs		
Increased probability of VLW	22.2	
Labour market returns to education	86.1	Behrman and Rosenzweig (2004)
Increased infant mortality	1,573.4	
Total	**1,681.7**	
Benefits		
Medical care per day: 855 USD		Alvear et al. (2013)
Year 2: ↑0.2 days	-171.0	
Year 3: 0.5 days	427.5	
Year 4: 0.4 days	342.0	
Year 5: 0.1 days	85.5	
PV at birth	562.5	
PV at mother's birth	39.9	
Welfare cost of taxation	19.9	García et al. (2020)
Total	**59.8**	

Notes: Each panel displays the costs and benefits in terms of real costs in 2022. Panel A displays the details for generation 1 and panel B for generation 2 considering 0.24 additional births up to 25 years. All costs are presented discounted to the time at birth of the first generation mother. In the case of benefits, these are reported in terms of time accrued, and then discounted to the time of birth of the mother in 'PV at mother's birth'.

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